2 OUTCOME-BASED TEACHING LEARNING (120)

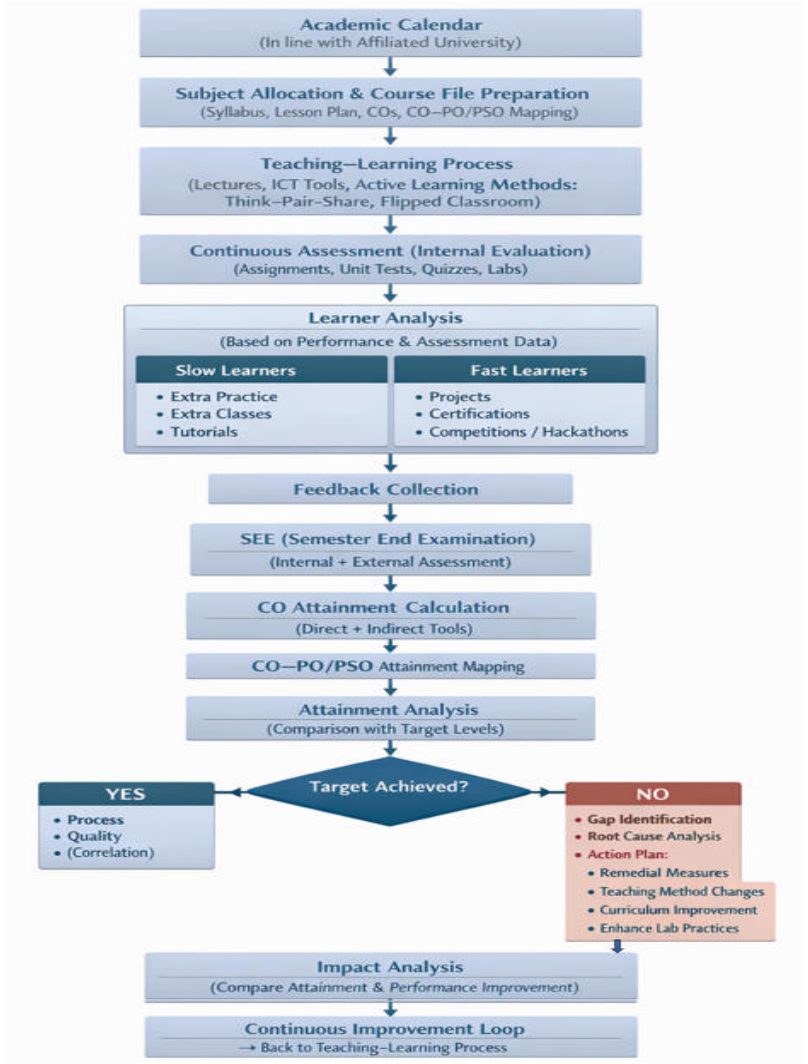
Total Marks 120.00

2.1 Describe Processes Followed to Ensure Quality of Teaching & Learning (20)

Total Marks 20.00

Institute Marks : 20.00

Teaching,Learning and Evaluation Process:



2.1.1 Academic Calendar

Academic Calendar Preparation

At the beginning of each academic year, the institution prepares a comprehensive semester-wise academic calendar in alignment with Affiliated University guidelines and Outcome-Based Education (OBE) practices. Each semester typically consists of:

- 15 weeks of instructional period, ensuring adequate coverage of curriculum and attainment of Course Outcomes (COs).
- 2 weeks allocated for Continuous Internal Evaluation (CIE), including mid-term examinations, quizzes, assignments, and other assessment components.

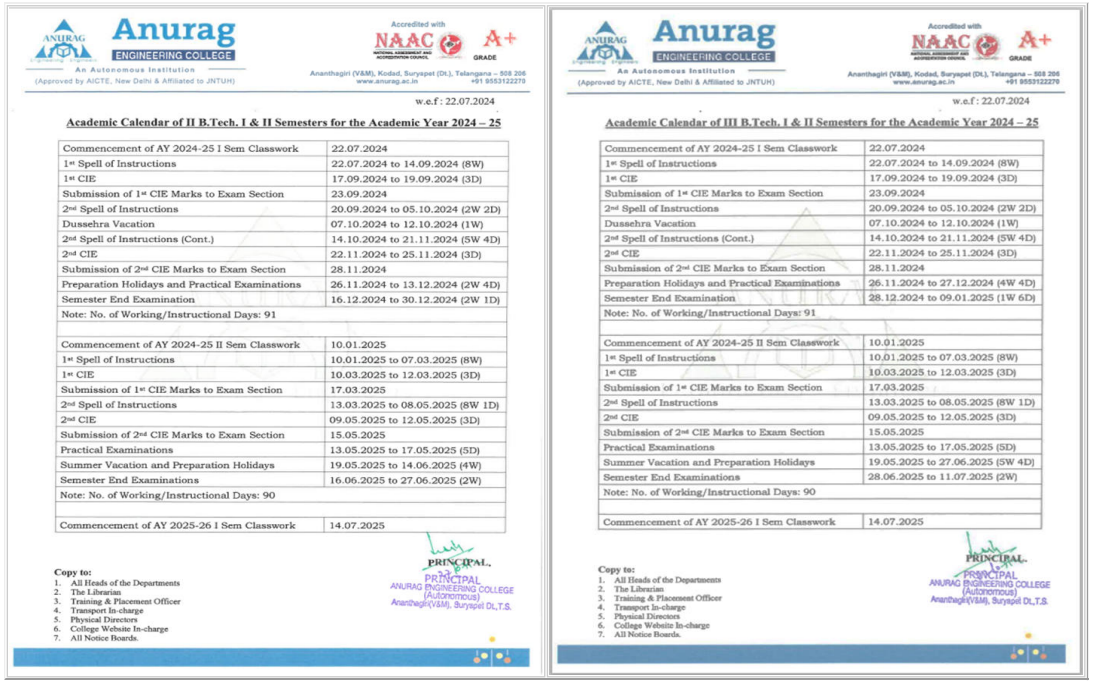
At the department level, a detailed academic calendar is prepared in synchronization with the institutional calendar. This departmental calendar ensures:

- Alignment with COs, POs, and PSOs, supporting systematic attainment tracking.
- Structured scheduling of theory and laboratory sessions, ensuring balanced delivery of conceptual and practical knowledge.
- Integration of discipline-specific activities, such as:
  - Laboratory experiments and project work
  - Skill development sessions
  - Guest lectures, workshops, and seminars

- Industrial visits and internships

- Provision for remedial classes and advanced learning, addressing diverse student learning needs.
- Continuous monitoring and review mechanisms, ensuring adherence to the plan and enabling corrective actions if required.

This structured planning ensures effective curriculum delivery, continuous assessment, and achievement of learning outcomes, in line with NBA accreditation requirements.



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An Autonomous Institution
(Approved by AICTE, New Delhi & Affiliated to JNTUH)
Accredited with NAAC A+ GRADE
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www.anurag.ac.in +91 9553122270

w.e.f : 22.07.2024

Academic Calendar of II B.Tech. I & II Semesters for the Academic Year 2024 – 25

Commencement of AY 2024-25 I Sem Classwork	22.07.2024
1st Spell of Instructions	22.07.2024 to 14.09.2024 (8W)
1st CIE	17.09.2024 to 19.09.2024 (3D)
Submission of 1st CIE Marks to Exam Section	23.09.2024
2nd Spell of Instructions	20.09.2024 to 05.10.2024 (2W 2D)
Dussehra Vacation	07.10.2024 to 12.10.2024 (1W)
2nd Spell of Instructions (Cont.)	14.10.2024 to 21.11.2024 (5W 4D)
2nd CIE	22.11.2024 to 25.11.2024 (3D)
Submission of 2nd CIE Marks to Exam Section	28.11.2024
Preparation Holidays and Practical Examinations	26.11.2024 to 13.12.2024 (2W 4D)
Semester End Examination	16.12.2024 to 30.12.2024 (2W 1D)
Note: No. of Working/Instructional Days: 91	
Commencement of AY 2024-25 II Sem Classwork	10.01.2025
1st Spell of Instructions	10.01.2025 to 07.03.2025 (8W)
1st CIE	10.03.2025 to 12.03.2025 (3D)
Submission of 1st CIE Marks to Exam Section	17.03.2025
2nd Spell of Instructions	13.03.2025 to 08.05.2025 (8W 1D)
2nd CIE	09.05.2025 to 12.05.2025 (3D)
Submission of 2nd CIE Marks to Exam Section	15.05.2025
Practical Examinations	13.05.2025 to 17.05.2025 (5D)
Summer Vacation and Preparation Holidays	19.05.2025 to 14.06.2025 (4W)
Semester End Examinations	16.06.2025 to 27.06.2025 (2W)
Note: No. of Working/Instructional Days: 90	
Commencement of AY 2025-26 I Sem Classwork	14.07.2025

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w.e.f : 22.07.2024

Academic Calendar of III B.Tech. I & II Semesters for the Academic Year 2024 – 25

Commencement of AY 2024-25 I Sem Classwork	22.07.2024
1st Spell of Instructions	22.07.2024 to 14.09.2024 (8W)
1st CIE	17.09.2024 to 19.09.2024 (3D)
Submission of 1st CIE Marks to Exam Section	23.09.2024
2nd Spell of Instructions	20.09.2024 to 05.10.2024 (2W 2D)
Dussehra Vacation	07.10.2024 to 12.10.2024 (1W)
2nd Spell of Instructions (Cont.)	14.10.2024 to 21.11.2024 (5W 4D)
2nd CIE	22.11.2024 to 25.11.2024 (3D)
Submission of 2nd CIE Marks to Exam Section	28.11.2024
Preparation Holidays and Practical Examinations	26.11.2024 to 27.12.2024 (4W 4D)
Semester End Examination	28.12.2024 to 09.01.2025 (1W 6D)
Note: No. of Working/Instructional Days: 91	
Commencement of AY 2024-25 II Sem Classwork	10.01.2025
1st Spell of Instructions	10.01.2025 to 07.03.2025 (8W)
1st CIE	10.03.2025 to 12.03.2025 (3D)
Submission of 1st CIE Marks to Exam Section	17.03.2025
2nd Spell of Instructions	13.03.2025 to 08.05.2025 (8W 1D)
2nd CIE	09.05.2025 to 12.05.2025 (3D)
Submission of 2nd CIE Marks to Exam Section	15.05.2025
Practical Examinations	13.05.2025 to 17.05.2025 (5D)
Summer Vacation and Preparation Holidays	19.05.2025 to 27.06.2025 (5W 4D)
Semester End Examinations	28.06.2025 to 11.07.2025 (2W)
Note: No. of Working/Instructional Days: 90	
Commencement of AY 2025-26 I Sem Classwork	14.07.2025

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2.1.2 Instruction Methods

Use of various instructional methods and pedagogical initiatives for enhancing Learning Experiences
The conventional chalk-n-talk teaching method is commonly adopted by all the teachers. It facilitates the teacher to interpret, explain and revise the content of a text only for better understanding of the subject by the learners. At the end of instruction of each unit, the students are given with specific assignments which enrich their learning.
ICT Enabled Teaching:
ICT enabled teaching methods have been made available in the institute with Wi-Fi facility. The Teaching- Learning Process is supported with Regular Practical Sessions, access to Digital Library, Online Courses like NPTEL and online journals. Use of LCD projectors for seminars and workshops, educational videos, communication skills training facility make the students to acquire proficiency in listening, speaking, reading and writing.



Participative Learning:
In participative learning methods, students are encouraged to participate in Project Competitions, Design Contests and various technical events.



Learner–centered Teaching:

In learner–centered teaching methods such as role play, project work, field visit, case study etc. supplement classroom teaching which enhances critical thinking among students.



The department also promotes industry sponsored projects and research projects.The students are encouraged to participate in project competitions, design contests, various technical festivals and arranging industrial visits in activity based learning.

Project Based Learning:

Project Based Learning is a teaching method in which students gain knowledge and skills by working for an extended period of time to investigate and respond to an authentic, engaging, and complex question, problem, or challenge.

The benefits of Project-Based Learning:

- Collaboration
- Problem Solving
- Creativity
- In-Depth Understanding
- Self-Confidence
- Critical Thinking
- Project Management
- Curiosity
- Empowerment

Digital Learning Management System (LMS):

The institution effectively integrates Google Classroom as a digital Learning Management System (LMS) to enhance the quality of teaching and learning processes. It serves as a centralized platform for academic delivery, continuous assessment, communication, and student engagement.

- **Blended Teaching–Learning Environment**

Google Classroom supports a blended learning model, enabling faculty to combine traditional classroom teaching with digital resources. Faculty upload:

- Lecture notes and PPTs
- Video lectures
- E-books and reference materials
- Assignments and project instructions

This ensures anytime, anywhere access to learning resources for students.

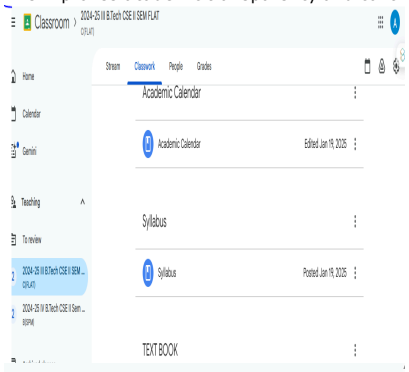
- **Structured Course Management**

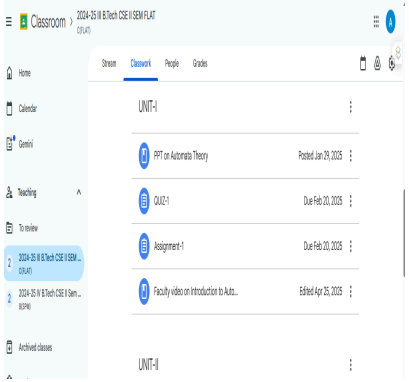
Each course is created as a virtual classroom where:

- Syllabi, Course schedules are uploaded

- Course modules are organized Unit - wise
- Important announcements and reminders are shared

This improves academic transparency and continuity.





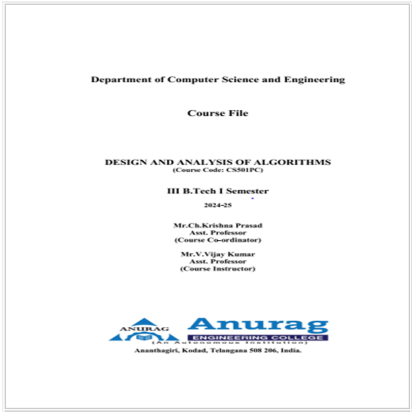
Collaborative Learning:

The Curriculum project work is made available to promote collaborative learning. Students are encouraged to form into groups with inter disciplinary combinations in addressing the real time problems. This leads to an eco-system making the students to learn by working together in collaboration.
Collaborative learning is an umbrella term for a variety of educational approaches involving joint intellectual effort by students, or students and teachers together. Usually students are working in groups of two or more, mutually searching for understanding, solutions or meanings, or creating a product.
Collaborative learning activities vary widely, but most center on students' exploration or application of the course material, not simply the teacher's presentation or explication of it".
The benefits of collaborative learning:

- Development of higher-level thinking, oral communication, self-management, and leadership skills.
- Promotion of student-faculty interaction.
- Increase in student retention, self-esteem, and responsibility.
- Exposure to and an increase in understanding of diverse perspectives.



Sample Course Files:



Department of Computer Science and Engineering

Course File

DESIGN AND ANALYSIS OF ALGORITHMS
(Course Code: CS501PC)

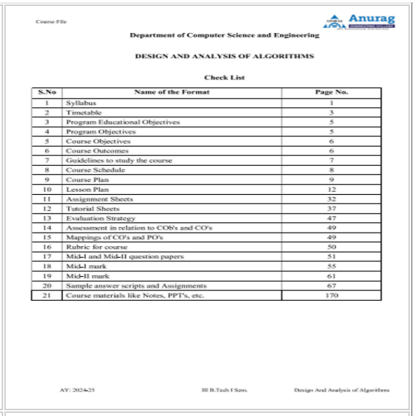
III B.Tech I Semester

2024-25

Mr.Ch.Krishna Prasad
Asst. Professor
(Course Co-ordinator)

Mr.V.Vijay Kumar
Asst. Professor
(Course Instructor)

Anurag Engineering College
(An Autonomous Institution)
Ananthagiri, Kodad, Telangana 508 206, India.



Course File — Anurag

Department of Computer Science and Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

Check List

S.No	Name of the Format	Page No.
1	Syllabus	1
2	Timetable	3
3	Program Educational Objectives	5
4	Program Objectives	5
5	Course Objectives	6
6	Course Outcomes	6
7	Guidelines to study the course	7
8	Course Schedule	8
9	Course Plan	9
10	Lesson Plan	12
11	Assignment Sheets	32
12	Tutorial Sheets	37
13	Evaluation Strategy	47
14	Assessment in relation to COb's and CO's	49
15	Mappings of CO's and PO's	49
16	Rubric for course	50
17	Mid-I and Mid-II question papers	51
18	Mid-I mark	55
19	Mid-II mark	61
20	Sample answer scripts and Assignments	67
21	Course materials like Notes, PPT's, etc.	170

AY: 2024-25 III B.Tech I Sem. Design And Analysis of Algorithms



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Department of Computer Science and Engineering

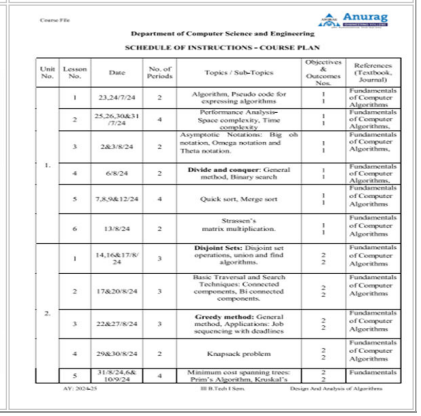
COURSE SCHEDULE

The Schedule for the whole Course / Subject is:

S. No.	Description	Duration (Date) From	Duration (Date) To	Total No. of Periods
1.	**UNIT-I** **Introduction:** Algorithm, Pseudo code for expressing algorithms, Performance Analysis--Space complexity, Time complexity, Asymptotic Notations: Big oh notation, Omega notation and Theta notation. **Divide and conquer**: General method, Applications: Binary search, Quick sort, Merge sort, Strassen's matrix multiplication.	23/07/24	13/08/24	16
2.	UNIT-II **Disjoint Sets:** Disjoint set operations, union and find algorithms. Basic Traversal and Search Techniques: Connected components, Bi connected components. **Greedy method:** General method, Applications: Job sequencing with deadlines, Knapsack problem, Minimum cost spanning trees: Prim's Algorithm, Kruskal's Algorithm.	14/08/24	10/09/24	15
3.	**UNIT-III:** **Dynamic Programming**: General method, applications: Optimal binary search tree, 0/1 knapsack problem, All pairs shortest path problem, Traveling salesperson problem.	11/09/24	1/10/24	10
4.	**UNIT-IV:** **Backtracking:** General method, Applications: n-queen's problem, Sum of subsets problem, Graph coloring. **Branch and Bound:** General method, 0/1 knapsack problem: LC Branch and Bound solution, FIFO Branch and Bound solution.	3/10/24	1/11/24	13
5.	**UNIT-V:** NP-Hard and NP-Complete problems: Basic concepts: Deterministic and Non-deterministic algorithms, NP-Hard and NP-Complete classes, Node cover Decision Problem,.	5/11/24	21/11/24	9

Total No. of Instructional periods available for the course: **63** Hours

AY: 2024-25 III B.Tech I Sem. Design And Analysis of Algorithms



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Department of Computer Science and Engineering

SCHEDULE OF INSTRUCTIONS - COURSE PLAN

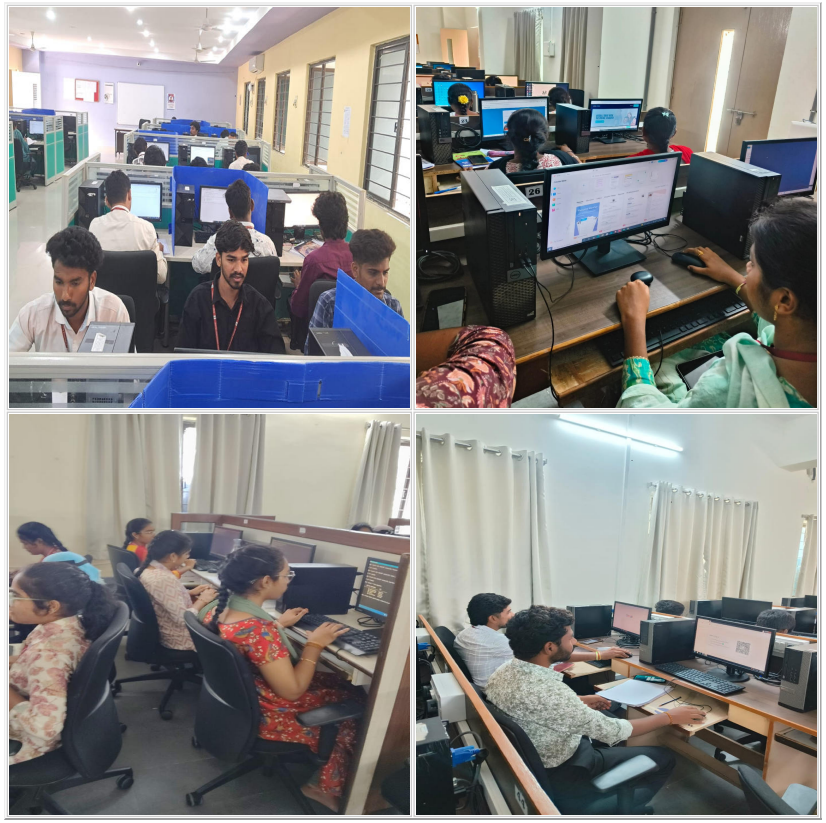
Unit No.	Lesson No.	Date	No. of Periods	Topics / Sub-Topics	Objectives & Outcomes Nos.	References (Textbook, Journal)
1.	1	23,24/7/24	2	Algorithm, Pseudo code for expressing algorithms	1 1	Fundamentals of Computer Algorithms
	2	25,26,30&31 /7/24	4	Performance Analysis--Space complexity, Time complexity.	1 1	Fundamentals of Computer Algorithms,
	3	2&3/8/24	2	Asymptotic Notations: Big oh notation, Omega notation and Theta notation.	1 1	Fundamentals of Computer Algorithms,
	4	6/8/24	2	**Divide and conquer**: General method, Binary search	1 1	Fundamentals of Computer Algorithms,
	5	7,8,9&12/24	4	Quick sort, Merge sort	1 1	Fundamentals of Computer Algorithms
	6	13/8/24	2	Strassen's matrix multiplication.	1 1	Fundamentals of Computer Algorithms
2.	1	14,16&17/8/24	3	**Disjoint Sets:** Disjoint set operations, union and find algorithms.	2 2	Fundamentals of Computer Algorithms
	2	17&20/8/24	3	Basic Traversal and Search Techniques: Connected components, Bi connected components.	2 2	Fundamentals of Computer Algorithms
	3	22&27/8/24	3	**Greedy method:** General method, Applications: Job sequencing with deadlines	2 2	Fundamentals of Computer Algorithms
	4	29&30/8/24	2	Knapsack problem	2 2	Fundamentals of Computer Algorithms
	5	31/8/24,6,& 10/9/24	4	Minimum cost spanning trees: Prim's Algorithm, Kruskal's	2 2	Fundamentals

AY: 2024-25 III B.Tech I Sem. Design And Analysis of Algorithms

2.1.3 Quality of Laboratory experience

The department has state-of-the-art laboratories for students to get their theoretical concepts clear through practical / experimental knowledge by creating an atmosphere where they are made to see and analyze everything in an objective and scientific way.

The department has latest software like Oracle, IBM Rational Rose, IBM Dream Spark etc., MoU with TASK ,IBM, Infosys etc.to give experimental knowledge to students on Data Science ,Artificial Intelligence and Machine Learning etc.

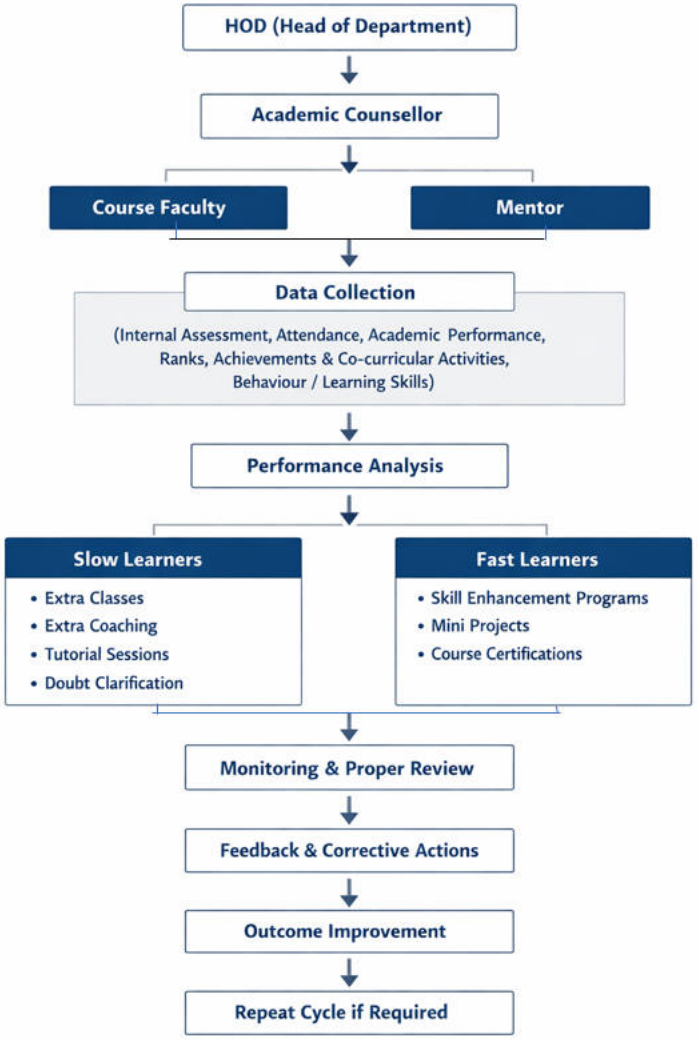


Impact analysis:

The following positive outcomes are observed after adopting the above mentioned innovative Teaching Learning Process

- Improved attendance of students in every class.
- Improvement in results.
- Active participation of students in OBE activities.
- Innovative project ideas are derived in class.
- Amicable bonding between students and faculty.
- Appreciation from the parents.

2.1.4-The institution assesses students' learning levels and organises special programmes for slow learners and fast learners



Encouraging Fast Learners and Assisting Slow Learners

Relevant student data is systematically collected, including internal assessment marks, attendance, academic performance, ranks, achievements, and behavioural aspects. This data is then analyzed to evaluate overall student performance.

Based on the analysis, students are classified into slow learners and fast learners. Slow learners are supported through remedial measures such as extra classes, coaching, tutorial sessions, and doubt clarification. Fast learners are encouraged through skill enhancement programs, mini projects, and course certifications.

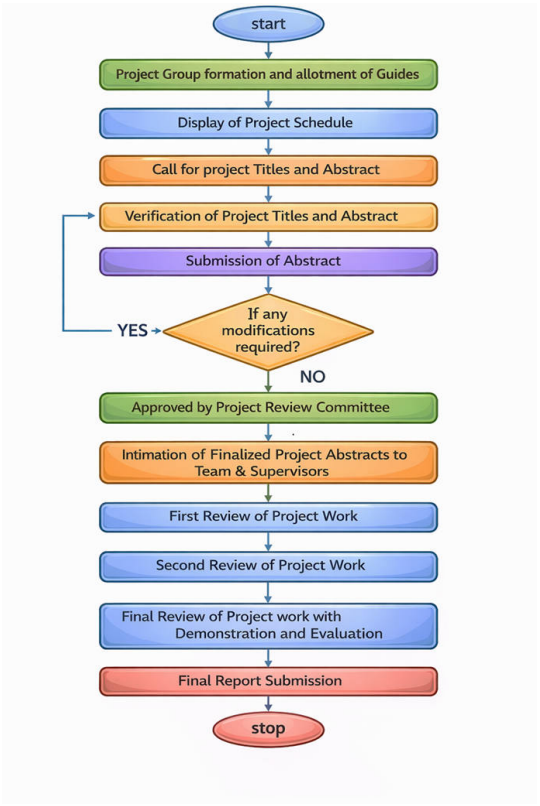
The process is followed by continuous monitoring and proper review to track progress. Feedback is collected and corrective actions are implemented to address identified gaps.

Finally, the process leads to outcome improvement, ensuring enhanced academic performance. This forms a continuous improvement cycle, which is repeated as required to maintain and improve the quality of teaching and learning.

2.2 Quality of Student Capstone Project (25)	Total Marks 25.00
	Institute Marks : 25.00

B.Tech Project Work shall be carried out in two stages: Project Stage – I during IV Year I Semester, Project Stage – II during IV Year II Semester.

Project Group: The students are formed into groups consisting of minimum three and maximum five members



A. Identification of Projects and allocation methodology to faculty members

- The guides are allocated based on their specialization or interest by the project coordinator and Head of the Department.
- The project work has two phases, on first phase the students are encouraged to identify the project based on their interest and also encouraged to do project work on real world problems
- During the first phase (Identification of Project), students collect the requirements, literature and design the workflow of the project. Similarly, the industrial projects are also planned and scheduled.
- Students implement, test the project ,analyze the results and submit the project reports in second phase.
- The project work could be evaluated by the reviews in proper intervals.

Sample Project Batch List:

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CSE-A PROJECT BATCHES

S.No	B.No	HT No	NAME	TITLE	SUPERVISOR	SUPERVISOR SIGNATURE
1	A1	21C11A0568	MADHURI SAMMIDI	Electronic Voting Systems Using Blockchain Technology.	Dr.M. Murugesan	
2		21C11A0502	AKHILA CHEKKA			
3		21C11A0526	DIVYA GOLI			
4		19C11A0552	MURUGESAN KEERTHI			
5		21C11A0507	ANJUM SHAIK			
6	A2	21C11A0519	BINDU PRASANNA GONGATI	Exploring cardiovascular disease risk through data analysis and machine learning models.	Dr. Y V R Naga Pawan	
7		21C11A0512	ANUSHA SURAGANI			
8		21C11A0511	ANUSHA NUNNA			
9		19C11A0505	PARA AKANKSHA			
10		21C11A0554	KIRANMAI NOMULA			
11	A3	21C11A0530	FATHIMA BEGUM SHAIK	Medicine Identification Application For Low Vision People	M.Anusha	
12		21C11A0569	MADHURI VORUGANTI			
13		21C11A0534	GNANASA KONDAPU REDDY			
14		21C11A0505	AFRA SHAIK			
15	A4	21C11A0509	ANKITHA GUNDA	Credit Card Fraud Detection using Machine Learning and Deep Learning Algorithms	CH. Krishna Prasad	
16		21C11A0504	AMEENA SHAIK			
17		21C11A0539	HARIKA YADAVELLI			
18		21C11A0506	AMREEN SULTHANA			

B. Types and relevance of the projects and their contribution towards attainment of POs & PSOs

Based upon the functional area of the projects, they are categorized as follows:

- Application oriented
- Product and process development
- Solution to the industrial specific problems

After categorizing the projects, they will be mapped with POs and PSOs and the attainments are assessed based on the following:

- Depth in fundamentals
- Clarity in problem analysis
- Impact on societal needs
- Methodology
- Modern tool usage
- Novelty of work
- Presentation and documentation
- Cost effectiveness and project management
- Future scope of the work
- Team wor

C.Process for monitoring and evaluation

Several reviews are conducted to monitor the progress and evaluate the project .

For **Project Stage-I**, the departmental committee consists of Head of the Department, project supervisor and a senior faculty member shall evaluate the project work for 75 marks and project supervisor shall evaluate for 25 marks.

Zeroeth review	The title of the project is tentatively fixed and the guides of the projects were allocated.

Stage-I First review	Problem Statement, Existing system, Proposed system and requirements will be reviewed.
Stage-I Second review	Workflow design will be reviewed.
Stage-I Evaluation	The methodology of the project will be reviewed.
Stage-II First review	Modules of the Project and Implementation of project will be reviewed.
Stage-II Second review	Testing, Execution and future scope of the project will be reviewed.
Stage-II Evaluation	Total Project work will be reviewed.

Rubrics for Evaluation:

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2024-25

CSE

RUBRICS FOR PROJECT STAGE-I (CS705PW):

- CO1: Able to identify a real-world research problem and articulate its scope and significance.
- CO2: Able to propose a feasible solution and differentiate it from existing approaches.
- CO3: Able to conduct a literature survey and establish technical & functional system requirements.
- CO4: Able to design the system architecture and decompose it into functional modules.
- CO5: Able to create design representations and data models for the proposed system.

PROJECT STAGE-I (CS705PW) EVALUATION:

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-I(25 Marks)					
	Criteria	Excellent (4)	Good (3)	Average (2)	Poor (1)
1	Problem Understanding	Clear, precise, well-defined problem and objectives	Minor gaps	Basic understanding	Unclear
2	Literature Survey	Multiple relevant sources, critical analysis	Adequate references	Few sources	No proper survey
3	Feasibility& Specification	Covers all aspects (technical, economic, operational)	Covers most	Partial	Missing
4	Presentation	Clear, confident, structured	Good	Basic	Poor

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-II (25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	Architecture	Clear, scalable, well-designed	Good structure	Basic	Poor
2	Modules	Well-defined with proper functionality	Mostly clear	Limited clarity	Poor
3	Design Representation	Proper diagrams (UML/DFD) with accuracy	Minor errors	Basic diagrams	Missing
4	Presentation	Clear explanation with demo	Good	Basic	Weak

INTERNAL EVALUATION BY DEPARTMENTAL COMMITTEE (75 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Problem Understanding & Objectives**	Clearly defines a real-world problem with strong justification; objectives are precise, measurable, and well-aligned	Problem is clear; objectives are relevant but lack full clarity/measurability	Problem partially defined; objectives vague or loosely connected	Problem unclear; objectives missing or irrelevant
2	**System Architecture & Design**	Well-structured architecture; clear modular design; detailed UML/DFD; efficient data models	Good design with modules and basic diagrams; minor gaps	Basic design present; lacks modular clarity or detailed diagrams	No clear architecture; incomplete or missing design
3	**Presentation**	Clear, confident delivery; well-organized slides; strong explanation and demo; answers confidently	Good presentation; minor issues in clarity or flow	Basic presentation; limited clarity and explanation	Poor, unorganized; unable to explain project
4	**Project Report**	Comprehensive, well-structured report; all sections included; clear results; error-free	Good report; minor issues in formatting/content	Incomplete or lacks depth; multiple issues	Poorly written; missing key sections
5	**Viva-Voce**	Thorough understanding; answers all questions accurately and confidently	Good understanding; answers most questions	Basic understanding; limited answers	Poor understanding; unable to answer

PROJECT STAGE-I (CS705PW):

CO-PO Mapping:

MAPPING	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11
CO-1						2	1	1	2	-	2
CO-2						2	2	1	2	1	2
CO-3						1	1	1	2	-	2
CO-4						2	1	2	2	3	2
CO-5						1	2	2	2	1	2

CO-PSO Mapping:

MAPPING	PSO-1	PSO-2
CO-1	1	1
CO-2	2	3
CO-3	2	2
CO-4	3	3
CO-5	3	3

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - I AWARD LIST

Year: IV **Semester:I** **Program : B.Tech** **Branch:CSE** **Academic Year : 2024 – 25**

Date(s) of Presentation: 16th November 2024

S.No.	H.T.No.	Name of the Student	Date of the Presentation	Title of the Project	Name of the Supervisor	Marks (Max: 25)	Marks (Max: 75)				Total Marks (100 M)	Sign.of the Proj. Supr.
						Proj. Supr. (25 M)	HoD (75 M)	Sr. Faculty (75 M)	Proj. Supr. (75 M)	Avg. of (8), (9) & (10)	(7) + (11)	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
1	17C11A05D6	Bathini Usha	16-11-2024	Bank Customers Churn Model	K.UpendraRao							
2	19C11A0505	Para Akanksha	16-11-2024	Effective Feature Engineering Techniques for heart disease prediction using machine learning	Dr. Y V R Naga Pawan							
3	19C11A0552	Murugesan Keerthi	16-11-2024	Electronic Voting Systems Using Blockchain Technology	Dr.M. Murugesan							
4	20C11A05E9	Sumanth Marisetty	16-11-2024	Tourist Place Reviews Sentiment Classification Using Machine Learning Techniques	K.Sai Kartheek							
5	21C11A0501	Ajay Satta	16-11-2024	AI for Environmental Monitoring and Prediction	N.Mownika							

Signature of the Senior Faculty
Name:

Signature of the HoD
Name: Dr Y V R Naga Pawan

- For **Project Stage-II**, the external examiner shall evaluate the project work for 75 marks and the project supervisor shall evaluate it for 25 marks.

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CSE

2024-25

RUBRICS FOR PROJECT STAGE-II (CS802PW):

- CO1: Able to select appropriate technologies and implement the designed solution through code
- CO2: Able to plan and execute systematic testing using defined methodologies and test cases
- CO3: Able to analyze, interpret, and present results demonstrating solution effectiveness
- CO4: Able to plan, implement, and execute project deployment in a real or simulated environment
- CO5: Able to write an effective technical report and demonstrate the complete project through presentation

PROJECT STAGE-II (CS802PW) EVALUATION:

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-I(25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Implementation**	Complete implementation with correct logic and innovation	Mostly correct with minor issues	Partial implementation	Incomplete / incorrect
2	**Testing**	Thorough testing with multiple cases and validation	Adequate testing	Limited testing	No proper testing
3	**Presentation**	Very clear, structured, confident explanation	Good clarity	Basic explanation	Poor communication

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-II (25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Results**	Accurate results with proper analysis and interpretation	Correct results with minor gaps	Partial results	Incorrect / no results
2	**Conclusion &Future Scope**	Conclusion clearly summarizes objectives, methodology, and key results; strong insights; future scope is realistic, innovative, and technically feasible	Conclusion covers major points; future scope is relevant but lacks depth	Basic summary with limited linkage to results; future scope is generic	Weak or missing conclusion; no meaningful future scope
3	**Presentation**	Excellent delivery with strong understanding	Good delivery	Basic explanation	Weak presentation

EXTERNAL EVALUATION BY EXTERNAL EXAMINER (75 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Presentation**	Clear, well-structured, confident delivery with strong visuals and complete understanding	Good clarity and organization, minor gaps	Basic explanation with limited clarity	Poor communication, unclear or unorganized
2	**Execution**	Fully functional project with correct logic, innovation, and efficient implementation	Mostly functional with minor issues	Partially working project	Incomplete or non-working
3	**Results**	Accurate results with deep analysis, interpretation, and justification	Correct results with some analysis	Limited or basic interpretation	Incorrect or missing results
4	**Project Report**	Comprehensive, well-structured, error-free, with diagrams, references, and proper formatting	Good documentation with minor issues	Basic report with missing details	Poor or incomplete report
5	**Viva-Voce**	Excellent subject knowledge, answers confidently and accurately	Good understanding with minor errors	Basic understanding	Unable to answer questions

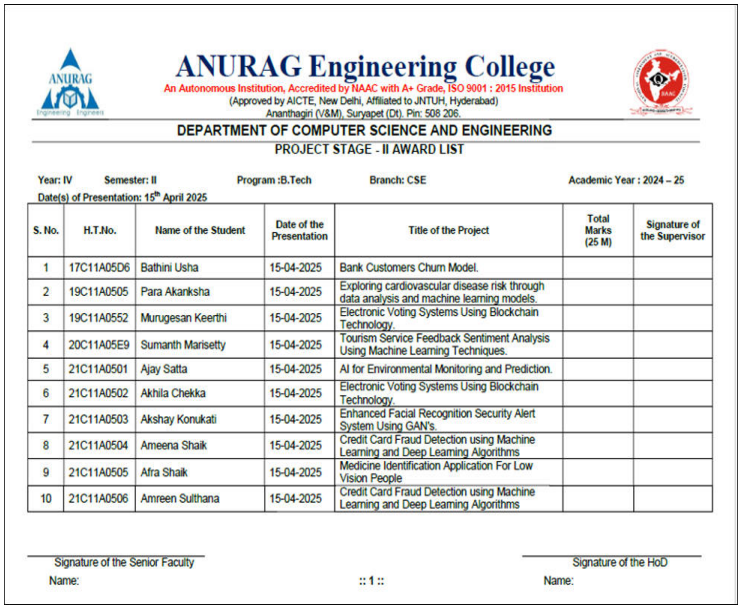
PROJECT STAGE-II (CS802PW):

CO-PO Mapping:

MAPPING	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11
CO-1						2	1	1	1	2	3
CO-2						2	2	1	1	2	2
CO-3						1	1	1	3	1	3
CO-4						2	1	2	2	3	2
CO-5						1	2	2	3	2	2

CO-PSO Mapping:

MAPPING	PSO-1	PSO-2
CO-1	3	3
CO-2	3	3
CO-3	2	2
CO-4	3	3
CO-5	3	2



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - II AWARD LIST

Year: IV Semester: II Program :B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation: 15th April 2025

S. No.	H.T.No.	Name of the Student	Date of the Presentation	Title of the Project	Total Marks (25 M)	Signature of the Supervisor
1	17C11A05D6	Bathini Usha	15-04-2025	Bank Customers Churn Model.		
2	19C11A0505	Para Akanksha	15-04-2025	Exploring cardiovascular disease risk through data analysis and machine learning models.		
3	19C11A0552	Murugesan Keerthi	15-04-2025	Electronic Voting Systems Using Blockchain Technology.		
4	20C11A05E9	Sumanth Marisetty	15-04-2025	Tourism Service Feedback Sentiment Analysis Using Machine Learning Techniques.		
5	21C11A0501	Ajay Satta	15-04-2025	AI for Environmental Monitoring and Prediction.		
6	21C11A0502	Akhila Chekka	15-04-2025	Electronic Voting Systems Using Blockchain Technology.		
7	21C11A0503	Akshay Konukati	15-04-2025	Enhanced Facial Recognition Security Alert System Using GAN's.		
8	21C11A0504	Ameena Shaik	15-04-2025	Credit Card Fraud Detection using Machine Learning and Deep Learning Algorithms		
9	21C11A0505	Afra Shaik	15-04-2025	Medicine Identification Application For Low Vision People		
10	21C11A0506	Amreen Sulthana	15-04-2025	Credit Card Fraud Detection using Machine Learning and Deep Learning Algorithms		

Signature of the Senior Faculty
Name:

:: 1 ::

Signature of the HoD
Name:

C. Process to assess individual and team performance

The performance of the individual team member of the project is assessed at the time of presentation in reviews by considering the following criterions:

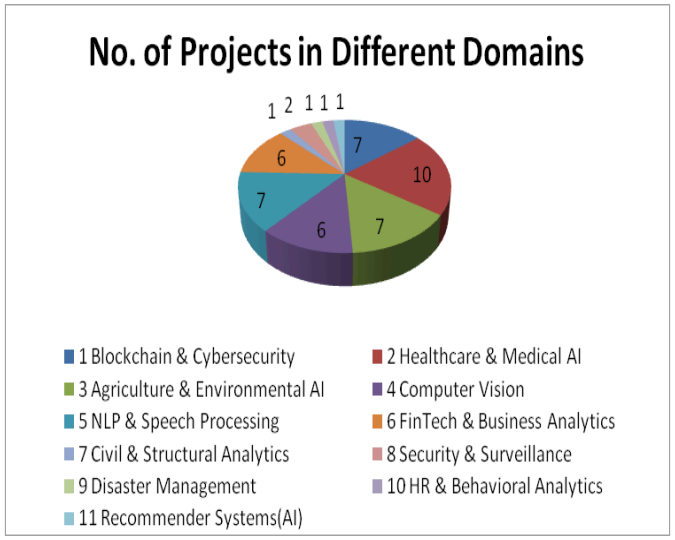
- Knowledge on the Project work
- Role in the project work
- Communication
- Competency

The performance of the project team is assessed by considering the following criterions:

- Knowledge of the each member and their contribution towards the project
- Coordination in deploying the work
- Schedule management

Student Projects and Related domains (AY.2024-25)

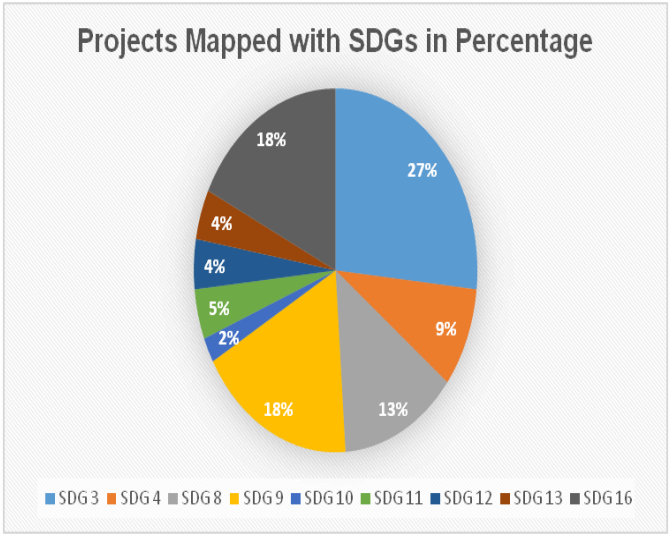
S.No	Domain	No. of Projects	Total Projects
1	Blockchain Technology	7	49
2	Computer Vision	6	
3	Machine Learning	7	
4	Big Data Analytics	9	
5	Cyber security	2	
6	Artificial Intelligence	18	



Projects mapped with SDGs:

SDG	Total No. of Projects Mapped
SDG 2	3
SDG 3	12
SDG 4	4
SDG 8	6
SDG 9	8
SDG 10	1
SDG 11	2
SDG 12	2
SDG 13	2
SDG 16	8

Graph Representing Projects Mapped with SDGs in Percentages:

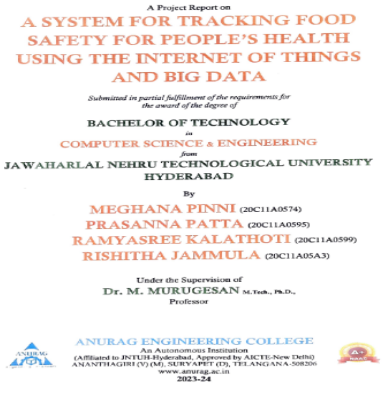


Best Projects Selection Process:

The best projects are selected based on the following metrics

1. Novelty of the Project
2. Understanding of the problem
3. Presentation of the solution
4. Knowledge of the problem
5. Domain
6. Technology specific to the problem domain
7. Analysis, interpretation and validity of results
8. Team work

S. No	Name of the Project	Team Members	Guide Name	Dept
1	A system for tracking food safety for peoples health using the Internet of Things and big data	Rishitha Jammula Meghana Pinni Ramyasree Kalathoti Prasanna Patta	Dr.M.MURUGESAN	CSE
2	Classification of Hyper spectral images	Arun Kumar Didigam Khurshid Ahammad Shaik Lokesh Domala Gopi Bangla	Dr. YVR NAGA PAWAN	CSE
3	A Machine Learning Approach For Tracking And Predicting Student Performance In Degree Programs	Sneha Reddy Annam Meenakshi Gudipati Manasa Mamidi Sai Kavya Pasupuleti	CH. KRISHNA PRASAD	CSE
4	A Machine Learning Based Lightweight Intrusion Detection System For The Internet Of Things	Rasheed Saida Shaik Saikrishna Maheshwarapu Nithin Sudheer N Ch Naga Venkata Sai koushik	G. SRINIVASA RAO	CSE
5	A Road Accident Prediction Model Using Data Mining Techniques	Tejasri Donthagani Vasanthi Vanama Vandana Baleboyina Sravani Tedla	P SANDEEP REDDY	CSE



A Project Report on

A SYSTEM FOR TRACKING FOOD SAFETY FOR PEOPLE'S HEALTH USING THE INTERNET OF THINGS AND BIG DATA

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

from

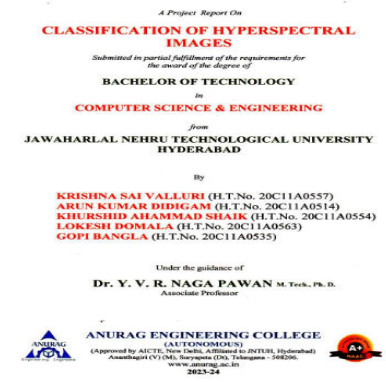
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

By

MEGHANA PINNI (20C11A0574)
PRASANNA PATTA (20C11A0595)
RAMYASREE KALATHOTI (20C11A0599)
RISHITHA JAMMULA (20C11A05A3)

Under the Supervision of
Dr. M. MURUGESAN M.Tech., Ph.D.,
Professor

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www.anurag.ac.in
2023-24



A Project Report On

CLASSIFICATION OF HYPERSPECTRAL IMAGES

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

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COMPUTER SCIENCE & ENGINEERING

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

By

KRISHNA SAI VALLURI (H.T.No. 20C11A0557)
ARUN KUMAR DIDIGAM (H.T.No. 20C11A0514)
KHURSHID AHAMMAD SHAIK (H.T.No. 20C11A0554)
LOKESH DOMALA (H.T.No. 20C11A0563)
GOPI BANGLA (H.T.No. 20C11A0535)

Under the guidance of
Dr. Y. V. R. NAGA PAWAN M. Tech., Ph. D.
Associate Professor

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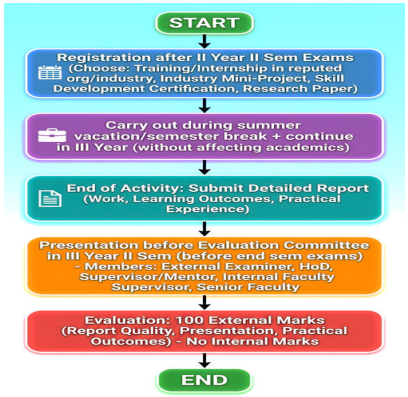
2.3 Internship/Industrial Training (10)	Total Marks 10.00

Institute Marks : 10.00

2.3.1 Internship:

The program encourages students to gain practical exposure to industry practices through Industry Training, Internship, Industry-Oriented Mini Projects, Skill Development Courses, or Research Paper Presentations in reputed journals related to their area of specialization. These activities help students connect theoretical knowledge with real-world applications and enhance their professional skills.

Process:



- Students are required to register for one of these activities immediately after the completion of the II Year II Semester examinations. The selected activity is carried out during the summer vacation or semester break and continued during the III Year, without affecting the regular academic schedule.
- The training or internship may be undertaken in reputed organizations or industries, while mini projects may be completed in collaboration with industries relevant to the student's specialization. Alternatively, students may enroll in skill development certification courses or prepare and present research papers in reputed journals, thereby strengthening their technical competence and research orientation.
- At the end of the activity, students must submit a detailed report documenting their work, learning outcomes, and practical experience. The report is then presented before an evaluation committee during the III Year II Semester, prior to the end semester examinations.
- The evaluation committee typically consists of:

1. An External Examiner
2. Head of the Department
3. Supervisor/Mentor of the Internship or Industry-Oriented Mini Project
4. Internal Faculty Supervisor
5. Senior Faculty Member of the Department

- The performance of the student is **evaluated for 100 external marks based on the quality of the report, presentation, and practical learning outcomes**. As per the academic regulations, **no internal marks are allotted** for this component.

This activity plays an important role in enhancing students' practical knowledge, industry exposure, research capability, and professional skills, thereby contributing significantly to the achievement of Program Outcomes related to problem solving, modern tool usage, and lifelong learning.

S. No	Activity	Duration / Time Period	Stage of Program	Evaluation
1	Industry Training	Summer Vacation / Semester Break (after II Year II Semester) and continuation during III Year	After II Year II Semester to III Year	100 External Marks
2	Internship in Reputed Organization	Summer Vacation / Semester Break and during III Year without	After II Year II Semester to III Year	100 External Marks

S. No	Activity	Duration / Time Period	Stage of Program	Evaluation
		affecting regular classes		
3	Skill Development Courses	Summer Vacation / Semester Break and during III Year	After II Year II Semester to III Year	100 External Marks
4	Paper Presentation in Reputed Journal	During III Year after registration in II Year II Semester	III Year	100 External Marks
5	Industry Oriented Mini Project	Conducted in collaboration with industry during III Year	III Year	100 External Marks
	Submission of Report and Presentation	Before End Semester Examination	III Year II Semester	Evaluated by External Committee

As **College Academic Council** suggested to perform the Internship activity from the above listed activities to enhance the students skills.

External Evaluation – 100 Marks

Conducted in III Year II Semester by External Examiner, HoD & Supervisor

Presentation	Industry Activity	Results & Discussion	Project Report	Viva-Voce
20 M	20 M	20 M	20 M	20 M

Evaluation Rubrics

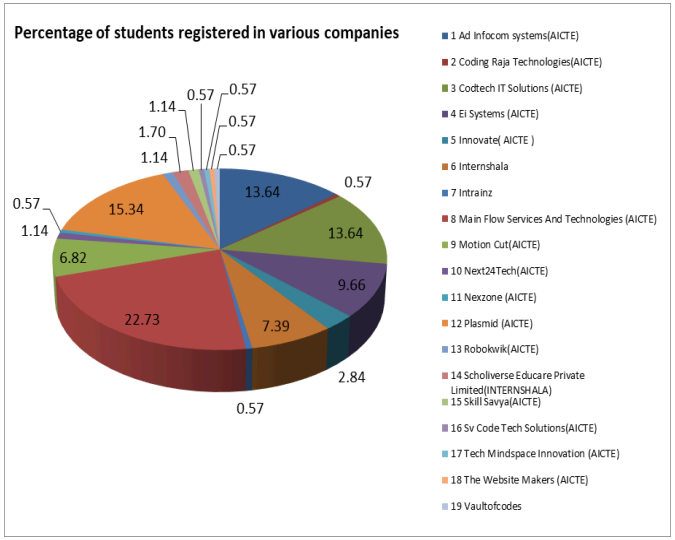
Criteria	Excellent (4)	Good (3)	Average (2)	Poor (1)
Presentation	Highly effective delivery, well-structured content, excellent visuals, confident communication	Clear delivery with minor issues, good structure and visuals	Basic presentation, some clarity issues, limited engagement	Poor delivery, unclear content, lack of preparation

Industry Activity	Strong industry relevance, active participation, excellent skill application, proper documentation	Good relevance and participation, adequate skills	Limited industry exposure, partial participation	No clear relevance, minimal involvement
Results and Discussion	In-depth analysis, accurate results, strong interpretation linked to objectives	Good analysis with minor gaps	Basic results with limited interpretation	Incorrect or missing analysis
Project Report	Well-organized, detailed, error-free, proper formatting and references	Mostly complete and organized with minor errors	Basic report with some missing elements	Poorly written, incomplete, lacks structure
Viva-Voce	Excellent subject knowledge, confident and accurate responses, strong reasoning	Good understanding, mostly correct answers	Basic knowledge, some difficulty answering	Poor understanding, unable to answer questions

List of different Industries where the students Registered for Internship:

SNO	Name of the Company	No. of Students Registered
1	Ad Infocom systems(AICTE)	24
2	Coding Raja Technologies(AICTE)	1
3	Codtech IT Solutions (AICTE)	24
4	Ei Systems (AICTE)	17
5	Innovate(AICTE)	5
6	Internshala	13
7	Intrainz	1
8	Main Flow Services And Technologies (AICTE)	40
9	Motion Cut(AICTE)	12
10	Next24Tech(AICTE)	2
11	Nexzone (AICTE)	1
12	Plasmid (AICTE)	27
13	Robokwik(AICTE)	2
14	Scholiverse Educare Private Limited(INTERNSHALA)	3
15	Skill Savya(AICTE)	2
16	Sv Code Tech Solutions(AICTE)	1
17	Tech Mindspace Innovation (AICTE)	1
18	The Website Makers (AICTE)	1
19	Vaultofcodes	1

Graph Representing the students Registered in Different Companies for Internship:



POs and PSOs Addressed:

SNO	Name of the Course	POs Addressed	PSOs Addressed
1	(CS606PW)-Industrial Oriented Mini Project/ Internship/Skill Development Course	PO6,PO7,PO8,PO9,PO10,PO11	PSO1,PSO2

The **Department of Computer Science and Engineering (CSE)** places strong emphasis on industry-oriented learning by encouraging students to undertake internships in reputed organizations. These internships serve as a platform for students to apply theoretical knowledge in real-world environments, particularly in domains such as **Software Development, Artificial intelligence, and Data-Driven applications**.

During the assessment period, **four CSE students** successfully completed internships in organizations including **Nextbyte, Medimind AI Solution, and Infovity**. The duration of internships ranged from **3 months to full-time engagement**, with stipends varying between **₹5,000 and ₹26,000 per month**.

List of Students went to Internship:

S.NO	HALL TICKET NUMBER	NAME	COMPANY	DURATION	STIPEND
1	22C11A0537	B. Geethika	Nextbyte	4 months	15,000
2	22C11A0556	SK. Khaja	Nextbyte	4 months	15,000
3	22C11A05E2	A. Siri Chandana	Medimind AI Solution	Full Time	26,000

S.NO	HALL TICKET NUMBER	NAME	COMPANY	DURATION	STIPEND
4	22C11A05I8	B. Yathin Karthikeya	Infovity	3 months	5,000

2.3.2 Industry Training:

Process:



- The Industry-Oriented Mini-Project / Summer Internship is to be completed during the vacation following the III Year II Semester examinations. Its evaluation will be carried out in the IV Year I Semester.
- Students must submit a detailed report of their work and present it before an evaluation committee. The presentation and report together carry 75 marks and will be assessed by a committee comprising an external examiner, the Head of the Department, and the project/internship supervisor.
- An additional 25 marks are allotted for internal assessment. These internal marks are awarded by the supervisor based on the student's performance and progress during the course of the Industry-Oriented Mini-Project / Summer Internship.

Evaluation Process:

Internal Evaluation – 25 Marks*(Awarded by Supervisor)*

Identification of Problem Domain and Detailed Analysis of the Existing Systems and Proposed Work	Description of Technical Details and Design Methodology	Project Implementation and Deployment.	Planning of Project Work and Team Structure.	Presentation and Report
5 M	5 M	5 M	5 M	5 M

External Evaluation – 75 Marks

Conducted in IV Year I Semester by External Examiner, HoD & Supervisor

Presentation	Industry Activity	Results & Discussion	Project Report	Viva-Voce
15 M	15 M	15 M	20 M	10 M

Rubrics for Internal Evaluation:



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2024-25

CSE

(CS704PW)MINI PROJECT/INTERNSHIP COURSE OUTCOMES:

- CO1: Identify the problem by applying acquired knowledge.
- CO2: Analyze and design executable project modules after considering risks.
- CO3: Choose efficient tools for constructing project modules.
- CO4: Integrate all the modules through effective team work.
- CO5: Elaborate the completed task and compile the project report.

RUBRICS FOR MINI PROJECT/INTERNSHIP EVALUATION :

INTERNAL EVALUATION BY SUPERVISOR # REVIEW-I (10 Marks)					
		Excellent (4)	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Identification of Problem Domain and Detailed Analysis of the Existing Systems and Proposed Work (CO1)**	Detailed and extensive explanation of the purpose and need of the project. Detailed explanation of the limitations of the existing systems. All objectives of the proposed work are well defined	Good explanation of the purpose and need of the project. Collects a great deal of information and good study of the existing systems. Good justification to the objectives	Average explanation of the purpose and need of the project. Moderate study of the existing systems. Incomplete justification to the objectives proposed	Minimal explanation of the purpose and need of the project. Minimal explanation of the specifications and the limitations of the existing systems. Objectives of the proposed work are either not identified or not well defined

2	**Description of Technical Details and Design Methodology (CO2)**	Strong description of the technical requirements of the project. Divison of problem into modules and good selection of computing framework Appropriate design methodology and properly justification	In-sufficient description of the technical requirements of the project. Divison of problem into modules and good selection of computing framework Design methodology not properly justified	In-sufficient description of the technical requirements of the project. Divison of problem into modules but design methodology not defined properly	Poor description of the technical requirements of the project. Modular approach not adopted. Design methodology not defined.

INTERNAL EVALUATION BY SUPERVISOR # REVIEW-II (15 Marks)					
		Excellent (4)	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Project Implementation and Deployment. (CO3)**	All defined objectives are achieved. Each module working well and properly demonstrated All modules of project are well integrated and system working is accurate.	All defined objectives are achieved. Each module working well and properly demonstrated. Integration of all modules not done and system working is not very satisfactory.	All defined objectives are achieved. Modules are working well in isolation and properly demonstrated. Modules of project are not properly integrated.	Defined objectives are not achieved. Modules are not in proper working form that further leads to failure of integrated system.
2	**Planning of Project Work and Team Structure. (CO4)**	Time frame properly specified and being followed Appropriate distribution of project work	Time frame properly specified and being followed Distribution of project work inappropriate	Time frame properly specified, but not being followed Distribution of project work un-even	Time frame not properly specified In-appropriate distribution of project work

External Evaluation Rubrics:

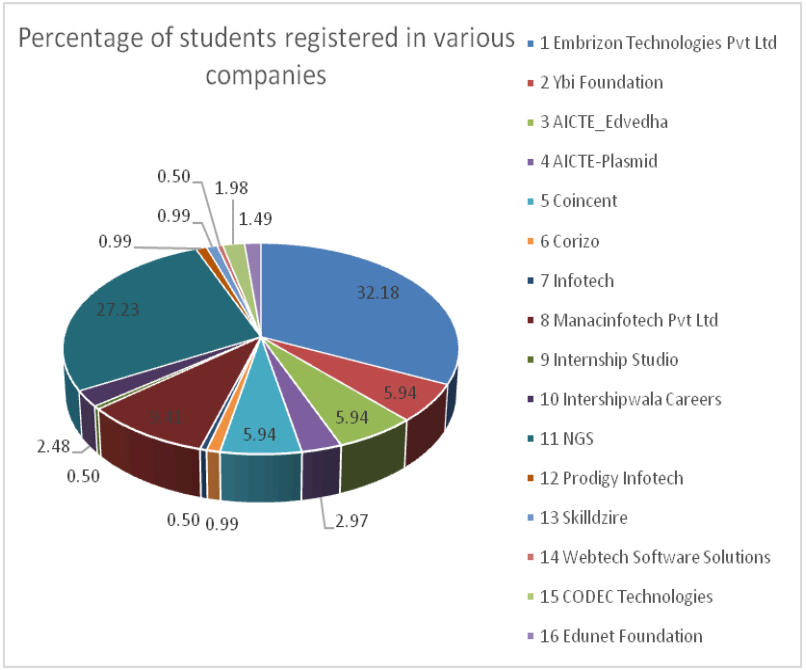
Criteria	Excellent (4)	Good (3)	Average (2)	Poor (1)
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Industry Activity	Strong industry relevance, active participation, excellent skill application, proper documentation	Good relevance and participation, adequate skills	Limited industry exposure, partial participation	No clear relevance, minimal involvement
Results and Discussion	In-depth analysis, accurate results, strong interpretation linked to objectives	Good analysis with minor gaps	Basic results with limited interpretation	Incorrect or missing analysis
Project Report	Well-organized, detailed, error-free, proper formatting and references	Mostly complete and organized with minor errors	Basic report with some missing elements	Poorly written, incomplete, lacks structure
Viva-Voce	Excellent subject knowledge, confident and accurate responses, strong reasoning	Good understanding, mostly correct answers	Basic knowledge, some difficulty answering	Poor understanding, unable to answer questions

List of different Industries where the students went for Industrial Training:

SNO	Name of the Company	No. of Students Registered
1	AICTE-Edvedha	12

SNO	Name of the Company	No. of Students Registered
2	AICTE-Plasmid	6
3	CODEC Technologies	4
4	Coincent	12
5	Corizo	2
6	Edunet Foundation	3
7	Embrizon Technologies Pvt Ltd	65
8	Infotech	1
9	Internship Studio	1
10	Intershipwala Careers	5
11	Manacinfotech Pvt Ltd	19
12	NGS	55
13	Prodigy Infotech	2
14	Skilldzire	2
15	Webtech Software Solutions	1
16	YBI Foundation	12

Graph Representing the No. of Students went to the Industrial Training in different Industries:



POs and PSOs Addressed:

SNO	Name of the Course	POs Addressed	PSOs Addressed
1	Industrial Visit	PO6,PO7,PO8,PO9,PO10,PO11	PSO1,PSO2

The Following Students who have successfully completed their internship at India Post Payments Bank, Suryapet.

S.NO	HALL TICKET NUMBER	NAME OF THE STUDENT	COMPANY	DURATION
1	21C11A0535	Neelam Gnanendra	India Post Payments Bank,Suryapet	28-08 -2024 to 27-09-2024
2	21C11A0590	Shaik Nazeem	India Post Payments Bank,Suryapet	28-08 -2024 to 27-09-2024
3	21C11A05A5	Rayapudi Praveen	India Post Payments Bank,Suryapet	28-08 -2024 to 27-09-2024

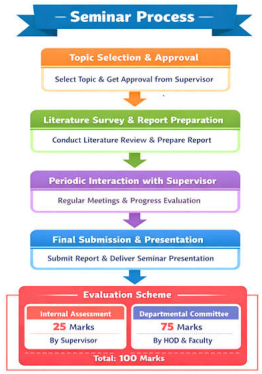
This internship opportunity provided students with practical knowledge and hands-on experience aligned with their academic curriculum. Training at a reputed organization like India Post Payments Bank enabled them to understand financial services, customer handling, and operational processes. Such industry exposure plays a vital role in enhancing technical competence, professional skills, and employability.

2.4 Seminar and Mini/Micro Projects (10) | **Total Marks 10.00**

Institute Marks : 10.00

2.4.1.SEMINAR

Process:



- **Topic Selection and Approval**
  At the beginning of IV Year II Semester, each student selects a topic relevant to their area of engineering. The Seminar Supervisor reviews and approves the topic to ensure it is technically sound, relevant to the curriculum, and aligned with current industry or research developments. Repetitive or superficial topics are avoided to maintain academic quality.
- **Literature Survey and Report Preparation**
  After approval, the student conducts a detailed literature survey using journals, conference proceedings, textbooks, and other credible sources. The emphasis is on understanding the subject thoroughly and presenting a clear analysis rather than reproducing existing material. Based on this study, the student prepares a structured technical report as per departmental guidelines, typically covering introduction, literature review, technical discussion, applications, future scope, and references.
- **Periodic Interaction with Supervisor**
  During the semester, the student meets regularly with the Seminar Supervisor for guidance and progress review. The supervisor evaluates the student's performance throughout this period and awards 25 internal marks based on regularity, initiative, understanding of the topic, and the quality of work submitted at different stages.
- **Final Submission and Presentation**
  The completed seminar report is submitted within the prescribed schedule. The student then delivers a formal presentation before a Departmental Committee consisting of the Head of the Department, the Seminar Supervisor, and a senior faculty member. The presentation is followed by a question-and-answer session to assess the student's depth of knowledge and clarity of understanding.
- **Evaluation Scheme**
  The seminar carries a total of 100 marks and does not include an external examination. The Supervisor awards 25 marks based on continuous assessment, while the Departmental Committee awards 75 marks based on the quality of the report and the effectiveness of the presentation. Evaluation parameters include technical content, presentation skills, response to questions, report quality, and adherence to prescribed guidelines and timelines.

Evaluation Process:

Internal Evaluation by Supervisor:

Criteria	Subject knowledge	Interactive Skills	Literature Review	Presentation Skills	Seminar Report	Total
Marks	5 M	5 M	5 M	5 M	5 M	25M

Internal Evaluation by Departmental Committee:

Criteria	Content	Speaking Skills& Audience interaction.	Visuals& Organization	Level of confidence	Closure	Total
Marks	15 M	15 M	15 M	15 M	15 M	75M

Rubrics for Seminar

ANURAG ENGINEERING COLLEGE

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(Affiliated to JNTU-Hyderabad, Approved by AICTE-New Delhi)
ANANTHAGIRI (V) (M), SURYAPETA (D), TELANGANA-508206
2024-25

CSE

RUBRICS FOR SEMINAR (CS801PW) - COURSE OUTCOMES (CO)

CO1: Evaluate recent topics of professional and personal interest.
CO2: Develop an ability to effectively present and interact with a group of people.
CO3: Respond respectfully to opposing ideas and develop ability to synthesize, evaluate and reflect on information.
CO4: Analyse the understanding of multidisciplinary concepts to apply theories, methods and knowledge to solve a problem.
CO5: Improve ability to utilise technical resources and to write seminar documents.

RUBRICS FOR SEMINAR (CS801PW) EVALUATION

INTERNAL EVALUATION BY SUPERVISOR (25 Marks)

		Excellent (4)	Good(3)	Average (2)	Poor (1)
1	**Subject knowledge (CO1)**	Demonstrates in-depth knowledge; answers questions with clear explanations and elaboration	Adequate knowledge of most topics; answers questions but lacks elaboration	Basic understanding; answers limited questions with difficulty	Does not have grasp of information; unable to answer questions
2	**Interactive Skills. (CO2)**	Exemplary body language, high confidence, strong eye contact, clear and loud voice	Good language skills and confidence; maintains eye contact and audible voice	Limited confidence; inconsistent eye contact and voice clarity	Poor body language; rarely maintains eye contact; not audible
3	**Literature Review. (CO3)**	Thorough and concise review; clearly connects previous work with current study	Describes related work and shows some connection to current work	Limited awareness of related work; weak connection to current work	Unaware of existing work; no context for current study
4	**Presentation Skills. (CO4)**	Excellent communication; effective use of multiple aids; completes within time	Moderate communication; satisfactory use of aids; minor time issues	Basic communication; limited use of aids; poor time management	Poor communication; ineffective presentation; fails to manage time
5	**Seminar Report. (CO5)**	Report strictly follows format; references and citations are accurate and well-presented	Report follows format with minor errors; limited references/citations	Report has format issues; insufficient references/citations	Report not as per format; references/citations inappropriate or missing

INTERNAL EVALUATION BY DEPARTMENTAL COMMITTEE (75 Marks)

		Excellent (4)	Good(3)	Average (2)	Poor(1)
1	**Content. (CO1)**	In-depth knowledge; main idea is clearly focused and supported with detailed information	Good knowledge; main idea is clear and supported with relevant information	Basic understanding; main idea somewhat clear with limited supporting details	Inadequate knowledge; main idea unclear and information is random
2	**Speaking Skills & Audience interaction. (CO2)**	All audience can hear; excellent eye contact; clear, expressive voice; confident posture	Most audience can hear; good eye contact; clear voice; minor nervousness	Some audience difficulty; limited eye contact; less clarity in voice; noticeable nervousness	Audience cannot hear; no eye contact; unclear/monotone voice; reads slides; lacks confidence
3	**Visuals & Organization. (CO3)**	Visually appealing; well-organized; logical flow; key points and conclusions clearly developed	Good layout; mostly organized; logical sequence with minor irrelevant content	Basic layout; some organization issues; key points not clearly emphasized	Poor layout; disorganized; difficult to follow; key points unclear
4	**Level of confidence. (CO4)**	Highly confident; clearly explains and answers queries with relevant examples	Good confidence; able to answer most queries with reasonable clarity	Moderate confidence; answers queries with limited clarity	Poor confidence; unable to answer queries effectively
5	**Closure. (CO5)**	Strong summary and impactful closing; captures audience attention effectively	Clear closing with good summary and audience engagement	Basic closing; limited summary; moderate audience engagement	Weak or no proper conclusion; lacks clarity and impact

POs and PSOs addressed to the Seminar:

SNO	Name of the Course	POs Addressed	PSOs Addressed

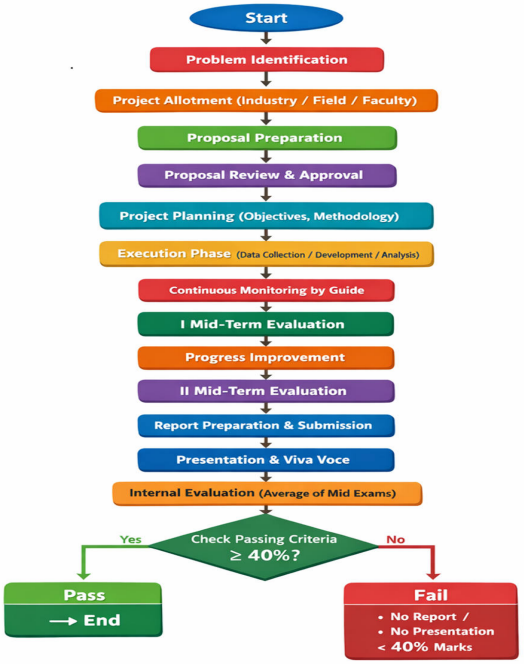
1.	(CS801PW)Seminar	PO6, PO8, PO9 ,PO10, PO11	PSO1,PSO2

2.4.2 Mini/Micro Projects

The Real-Time/Field-Based Research Project enables students to identify and solve real-world problems through a structured approach. The process begins with problem identification and project allotment, followed by proposal preparation and approval. Students then carry out project planning and execution with continuous guidance from faculty.

The progress is assessed through two mid-term evaluations, and students improve their work based on feedback. Finally, they submit a project report and give a presentation (viva voce) before an internal committee.

The final evaluation is based on the average of the two mid-term assessments, and students must secure at least 40% marks to pass. The process ensures development of technical, analytical, communication, and project management skills aligned with POs and PSOs.



- **Identification of Organization / Problem Statement**
  Students undertake the internship in an industry, research organization, or relevant institution, or complete a mini-project under faculty guidance. The work is based on real industry needs, practical engineering problems, or ongoing departmental research. Prior approval from the department is obtained before starting the internship or project to ensure relevance and academic value.
- **Execution During Vacation Period**
  The internship or mini-project is carried out during the vacation after the III Year II Semester examinations. During this period, students actively work on the assigned problem, interact with industry personnel where applicable, and apply engineering methods to develop solutions. Depending on the discipline, the work may involve design, development, testing, analysis, fabrication, or process study.
- **Supervision and Periodic Evaluation**
  Each student or group is assigned a faculty supervisor who monitors progress and provides guidance. The supervisor conducts periodic evaluations based on the student's involvement, progress, technical understanding, and professional conduct. These continuous assessments form the basis for awarding 25 marks for internal evaluation.
- **Report Preparation**
  After completion of the internship or mini-project, students prepare a detailed technical report describing the objectives, methodology, work carried out, results, and conclusions. The report reflects the student's understanding of the industrial context and the engineering problem addressed. It is prepared according to departmental guidelines and submitted at the beginning of IV Year I Semester.
- **Evaluation in IV Year I Semester**
  The submitted report and presentation are evaluated in IV Year I Semester by a committee consisting of an External Examiner, the Head of the Department, and the Project Supervisor. Students present their work and respond to questions from the committee. The committee awards 75 marks based on the quality of work, level of understanding, practical relevance, and presentation. Together with the 25 marks from internal assessment, the total evaluation is for 100 marks.

Evaluation Process:

Evaluation for MID - I

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REAL-TIME RESEARCH PROJECT / FIELD BASED PROJECT AWARD LIST

Year: II Semester: II **Program : B.Tech (CSE)** **MID: I** **Academic Year : 2023 – 24**

Date(s) of Presentation: ddth – ddth MMM 2024

S.NO	H.T.No	Student Name	Problem Identification	Literature Objectives	Work Program	Results	Presentation	Total
			10M	10M	10M	10M	10M	50 M

Evaluation for MID - II

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REAL-TIME RESEARCH PROJECT / FIELD BASED PROJECT AWARD LIST

Year: IV **Semester: II** **Program : B.Tech (CSE)** **MID: II** **Academic Year : 2023 – 24**

Date(s) of Presentation: ddth – ddth MMM 2024

S. No.	H.T. No	Student Name	Objectives Abstract & Work Program	Results	Presentation	Total
			20M	15M	15M	50 M

Final Evaluation

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REAL-TIME RESEARCH PROJECT / FIELD BASED PROJECT AWARD LIST

Year: II Semester: II Program : B.Tech (CSE) Academic Year : 2023 – 24
Max. Marks: 50

S. NO.	H.T.No	Student Name	Title of the Real-time / Field Project	MID – I	MID – II	Average

POs and PSOs Addresses to the Mini Project:

SNO	Name of the Course	POs Addressed	PSOs Addressed
1	Real-Time Project	PO6,PO7, PO8, PO9 ,PO10, PO11	PSO1,PSO2

2.5 Case Studies and Real-Life Examples (10) Total Marks 10.00

Institute Marks : 10.00

2.5.1 Case Studies:

The course incorporates case studies and real-life examples to bridge the gap between theoretical knowledge and practical application. These case studies are designed to expose students to real-world engineering problems and industry practices, enabling them to analyze situations, identify challenges, and develop suitable solutions.

Through case studies, students are encouraged to:

- Apply engineering knowledge to real-world problems
- Perform problem analysis and critical thinking
- Design and develop practical and innovative solutions
- Use modern tools and technologies relevant to the discipline
- Work collaboratively and improve communication and teamwork skills

Overall, the inclusion of case studies and real-life examples enhances **experiential learning, industry awareness, and practical competency**, thereby supporting the holistic development of students in alignment with the objectives of outcome-based education.

Type and Complexity

The case studies vary in type and complexity, ranging from conceptual understanding to system-level problem solving. They include examples from software development, system design, data analysis, and emerging technologies, allowing students to understand how engineering principles are applied in real-life scenarios.

The complexity of the case studies is structured to gradually enhance students' learning, beginning with basic application-oriented examples and progressing toward complex problem-solving scenarios that require analytical and design skills.

PO and PSO addressed:

These case studies contribute to the attainment of **Program Outcomes (POs)** such as **engineering knowledge, problem analysis, design and development of solutions, and use of modern tools**, along with **Program Specific Outcomes (PSOs)** related to the application of computing technologies for solving real-world problems.

Some of the Case Studies and Real Life Examples discussed in the courses are listed below

S.No	Name of the Course	Case Study / Real Life	Description of Case Study / Real Life Example	PO/PSO Addressed	Type & Complexity
1	DevOps Laboratory	CI/CD Pipeline for Food Delivery Application	Students design and implement an automated CI/CD pipeline for a food delivery web application similar to platforms like Zomato and Swiggy .The pipeline automatically builds, tests, and deploys the application ensuring scalability, reliability, and rapid delivery of software updates.	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PSO1, PSO2	Complex Engineering Problem

2.5.2 Real – Life Examples:

Real-life examples in Computer Science and Engineering involve the development of software-based solutions such as web and mobile applications to address practical problems in areas like healthcare, education, agriculture, safety, and productivity. These projects demonstrate the application of core CSE concepts including data structures, algorithms, artificial intelligence, database management, and software engineering principles. Students analyze real-world problems, design and implement efficient systems using modern tools and technologies, and evaluate their performance.

SNO	Team (Roll No – Name)	Real Life Example	Type	Complexity	PO/PSO Addressed
1	22C11A05E2–A. Siri Chandana, 22C11A05F7–ThotaSupriya, 22C11A05E3–V. Siri Thorani, G. Prashanthi	Predictive Personal Safety System for Women	Mobile App	Moderate–High	PO6, PO8, PO9, PSO1, PSO2

SNO	Team (Roll No – Name)	Real Life Example	Type	Complexity	PO/PSO Addressed
2	22C11A05F7–ThotaSupriya, 22C11A05F8–P. Surya, 22C11A05G0–B. Tejaswi, 22C11A05I0–B. Vijay, 22C11A05I6–A. VivekVardhan, 23C15A0511–P. Vijay Kumar	Task Timer & Productivity Tracker	Web App	Low–Moderate	PO8, PO9, PO10, PSO1, PSO2
3	22C11A05B5–M. Sahithi, 22C11A0583–K. Narendar, 22C11A0569–A. Mallika, D. Madhav Prakash, 22C11A0573–A. Manisha, 22C11A05B6–T. Sahitya	College Buddy Finder	Web App	Low–Moderate	PO6, PO8, PO9, PO10, PSO2
4	22C11A05D2–V. Sathwika, 22C11A0574–A. Manisha	Smart Anemia Care System	Web App	Moderate–High	PO6, PO7, PO9, PSO1, PSO2
5	22C11A0537–G. Bandaru, 22C11A0531–P. DurgaBhavani	Buddy Finder & Movie Recommendation	Web App	Low–Moderate	PO8, PO9, PSO1, PSO2
6	22C11A0537–G. Bandaru, 22C11A0516–Sumayyah, DurgaBhavani	AI Powered Smart Organic Farming	Mobile App	Moderate–High	PO6, PO7, PO11, PSO1, PSO2

2.6 SWAYAM/NPTEL/MOOC/Self Learning (10) **Total Marks 10.00**

Institute Marks : 10.00

SWAYAM, NPTEL, and other MOOC platforms provide opportunities for students to learn advanced topics beyond the prescribed curriculum through online self-learning. These platforms support flexible, accessible, and high-quality education developed by leading institutions and faculty.

1. SWAYAM Platform: SWAYAM (Study Webs of Active Learning for Young Aspiring Minds) is an initiative of the Government of India designed to achieve the principles of access, equity, and quality in education. It offers online courses for school, undergraduate, postgraduate, and professional learners.

Features:

- Free online courses from top universities and institutions
- Video lectures, reading materials, quizzes, and assignments
- Credit transfer facility as per University Grants Commission (UGC) and All India Council for Technical Education (AICTE) guidelines
- Certification after successful completion of exams

2. NPTEL Courses: National Programme on Technology Enhanced Learning (NPTEL) is a joint initiative of the Indian Institutes of Technology (IITs) and the Indian Institute of Science.It provides high-quality engineering, science, and management courses taught by expert faculty.

Key Benefits:

- Industry-relevant and advanced technical content
- Certification exams conducted nationally
- Courses aligned with emerging technologies such as AI, Data Science, IoT, Cybersecurity, etc.
- Helps students improve technical competency and employability

3. MOOC (Massive Open Online Courses):MOOCs are online courses designed for large-scale participation and open access through the internet. They allow students to learn from global universities and industry experts.

Characteristics:

- Flexible learning at any time and place
- Access to global learning resources
- Certification options available
- Enhances interdisciplinary know ledge

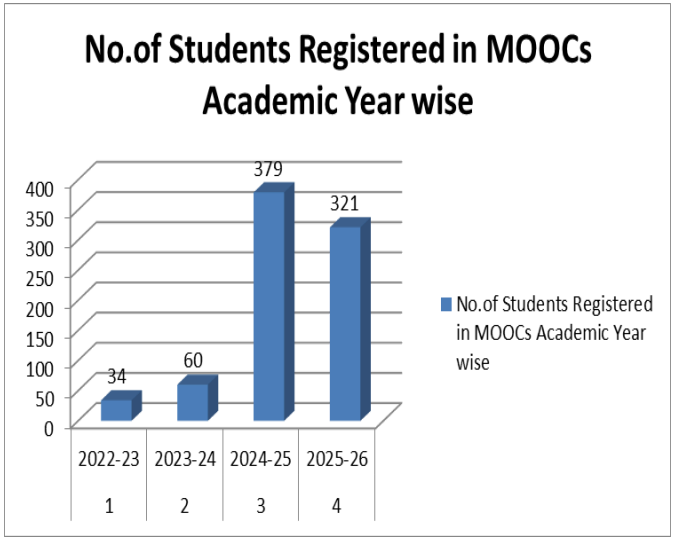
Examples include platforms like:

- Coursera
- edX
- Udemy

2.6.1 Number of students registered in MOOC Courses

SNO	Academic Year	No.of Students Registered
1	2022-23	34
2	2023-24	60
3	2024-25	379
4	2025-26	321

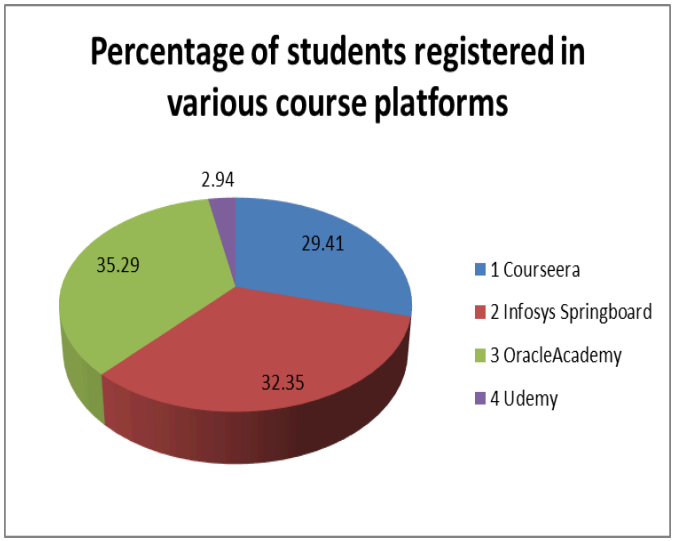
Graph Representation of Students Registered Academic Year wise:



Academic Year: 2022-23

SNO	Name of the Course Platform	No. of Students Registered
1	Courseera	10
2	Infosys Springboard	11
3	Oracle Academy	12
4	Udemy	1

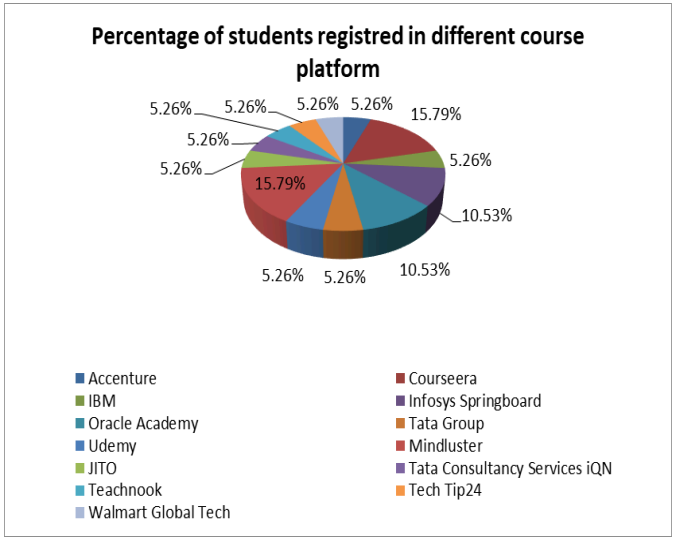
Graph Representation of Students Registered in different Course Platforms:



Academic Year: 2023-24

S.No	Name of the Course Platform	No. of Students Registered
1	Accenture	1
2	Courseera	15
3	IBM	1
4	Infosys Springboard	15
5	Oracle Academy	18
6	TATA	1
7	Udemy	1
8	Mindluster	3
9	JITO	1
10	TCS iQN	1
11	TEACHNOOK	1
12	TECH TIP24	1
13	Walmart Global Tech	1

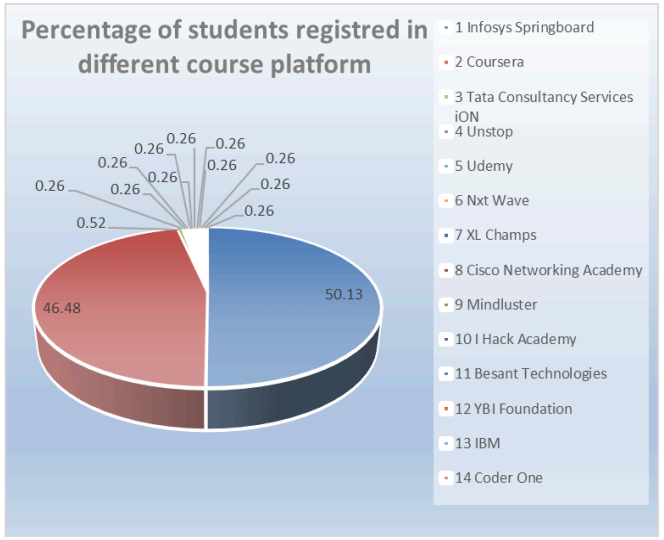
Graph Representation of Students Registered in different Course Platforms:



Academic Year: 2024-25

SNO	Name of the Course Platform	No. of Students Registered
1	Infosys Springboard	192
2	Coursera	178
3	Tata Consultancy Services iON	2
4	Unstop	1
5	Udemy	1
6	Nxt Wave	1
7	XL Champs	1
8	Cisco Networking Academy	1
9	Mindluster	1
10	I Hack Academy	1
11	Besant Technologies	1
12	YBI Foundation	1
13	IBM	1
14	Coder One	1

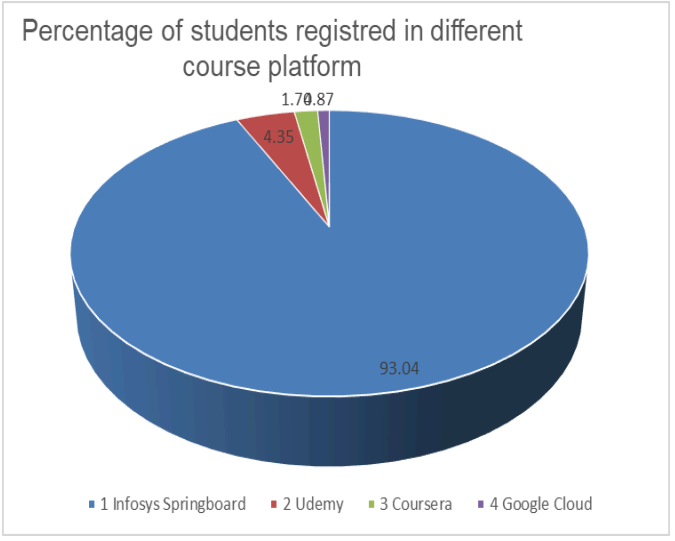
Graph Representation of Students Registerted in different Course Platforms:



Academic Year 2025-26:

SNO	Name of the Course Platform	No.Of Students Registered
1	INFOSYS SPRINGBOARD	321
2	UDEMY	15
3	COURSEERA	6
4	GOOGLE CLOUD	3

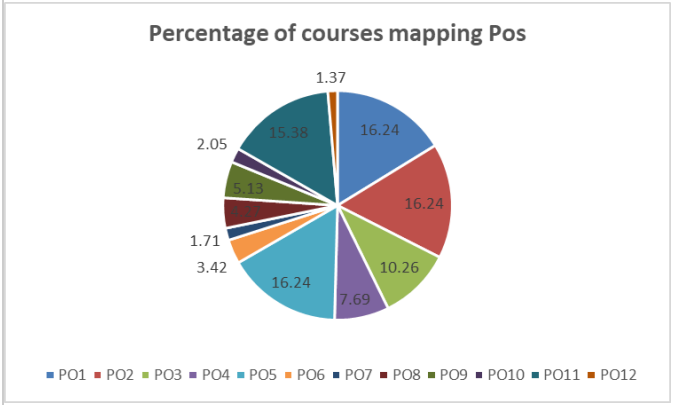
Graph Representation of Students Registered in different Course Platforms:



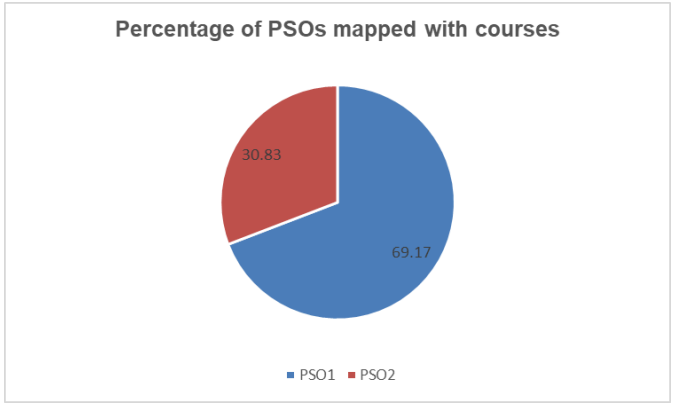
2.6.2 Certifications and PO/PSO addressed

Academic Year 2024-25:

Graph Representation of POs addressed to the Courses:



Graph Representation of PSOs addressed to the Courses:



2.7 Solving Complex Engineering Problems Incorporating Sustainability Goals (20) | **Total Marks 20.00**

Institute Marks : 20.00

The B.Tech (CSE) curriculum under R22 is designed with a structured and progressive approach to develop students' ability to solve complex engineering problems. It integrates foundational courses, core subjects, laboratories, and skill development activities across all four years. In the first year, emphasis is placed on building computational thinking, programming skills, engineering practices, and environmental awareness. The second year strengthens analytical and problem-solving abilities through core subjects such as Data Structures, Object-Oriented Programming, Operating Systems, Database Management Systems, and Software Engineering, along with real-time and societal projects that expose students to practical and community-oriented challenges.

In the third year, the curriculum advances into multidisciplinary and emerging technologies such as Machine Learning, Artificial Intelligence, Computer Networks, and DevOps, supported by industry-oriented mini projects and internships that provide hands-on experience in real-world problem solving. The final year culminates in capstone projects (Project Stage–I and II), where students integrate knowledge from various domains to design and implement solutions considering technical, economic, environmental, ethical, and societal constraints. Additionally, professional and open electives in areas like IoT, Cloud Computing, Cyber Security, and sustainability domains enhance multidisciplinary learning and align the program with relevant Sustainable Development Goals (SDGs), ensuring graduates are well-prepared to address complex engineering challenges.

Details of Core Courses mapped to SDGs:

Sem	Course Code	Course Title	How it supports	Likely SDGs
I-I	CS103ES	Programming for Problem Solving	Develops logical thinking, algorithm design, and structured problem solving	SDG 4, 9
I-I	EG105ES	Computer Aided Engineering Graphics	Helps students visualize, model, and communicate design ideas	SDG 9
I-I	CS106ES	Elements of Computer Science & Engineering	Introduces computing systems and basic engineering problem context	SDG 4, 9
I-I	CS109ES	Programming for Problem Solving Laboratory	Converts logic into working programs; builds implementation skill	SDG 4, 9
I-I	EE104ES	Basic Electrical Engineering	Builds engineering fundamentals for system understanding and energy awareness	SDG 7, 9
I-I	EE108ES	Basic Electrical Engineering Laboratory	Gives hands-on practice in measurement, circuits, and practical engineering skills	SDG 7, 9
I-I	HS110MC	Constitution of India	Builds social responsibility, ethics, and responsible citizenship	SDG 16
I-II	ME203ES	Engineering Workshop	Gives practical exposure to tools, fabrication, safety, and applied problem solving	SDG 8, 9
I-II	EC205ES	Electronic Devices and Circuits	Builds understanding of electronic components used in real systems	SDG 9
I-II	CS207ES	Python Programming Laboratory	Supports data handling, automation, prototyping, and computational solutions	SDG 4, 9
I-II	CS209ES	IT Workshop	Improves digital productivity, software tool usage, and technical readiness	SDG 4, 9
I-II	ES210MC	Environmental Science	Directly creates awareness on environmental protection and sustainability	SDG 6, 12, 13
II-I	PS301BS	Probability and Statistics	Supports data-driven decision making, modelling, prediction, and uncertainty handling	SDG 4, 9
II-I	ED302ES	Digital Electronics	Builds foundation for logic design and embedded/system-level engineering problems	SDG 9

II-I	CS303PC	Data Structures	Teaches efficient data organization and algorithmic problem solving	SDG 4, 9
II-I	CS304PC	Computer Organization and Architecture	Helps students understand system design, optimization, and resource constraints	SDG 9
II-I	CS305PC	Object Oriented Programming through Java	Supports modular design, reusability, and software solution development	SDG 4, 9
II-I	CS306PC	Data Structures Laboratory	Applies abstract data concepts to practical programming problems	SDG 4, 9
II-I	CS307PC	Object Oriented Programming through Java Laboratory	Builds hands-on software design and implementation ability	SDG 4, 9
II-I	CS308PC	Skill Development Course / Data Visualization – R Programming / Power BI	Helps students analyse, interpret, and present data for decision making	SDG 4, 9, 11
II-I	HS309MC	Gender Sensitization	Strengthens inclusion, equity, and social responsibility in engineering practice	SDG 5, 10, 16
II-II	CS402PC	Discrete Mathematics	Supports analytical reasoning used in algorithms, logic, and computing systems	SDG 4, 9
II-II	CS403PC	Operating Systems	Deals with resource allocation, concurrency, scheduling, and system efficiency	SDG 9
II-II	CS404PC	Database Management Systems	Supports structured data storage, retrieval, and scalable information systems	SDG 9, 11
II-II	CS405PC	Software Engineering	Teaches requirement analysis, design, testing, quality, and lifecycle thinking	SDG 9, 12
II-II	CS406PC	Operating Systems Laboratory	Applies scheduling, process handling, and system-level concepts in practice	SDG 9
II-II	CS407PC	Database Management Systems Laboratory	Builds practical skills in designing and implementing real database systems	SDG 9, 11
II-II	CS408PW	Real-time Research Project / Societal Related Project	Directly addresses real-life community or societal problems through engineering solutions	SDG 3, 4, 6, 9, 11, 13

II-II	CS409PC	Skill Development Course / NodeJS / React JS / Django	Supports application development, rapid prototyping, and deployment-oriented problem solving	SDG 4, 9
II-II	HS411MC	Intellectual Property Rights	Builds awareness on innovation, ethics, originality, and technology ownership	SDG 9, 16
III-I	CS501PC	Design and Analysis of Algorithms	Develops efficient solution strategies for complex computational problems	SDG 4, 9
III-I	CS502PC	Computer Networks	Supports communication systems, connectivity, and infrastructure-level problem solving	SDG 9, 11
III-I	CS503PC	DevOps	Introduces integration, deployment, automation, scalability, and reliability	SDG 9, 12
III-I	CS504PC	Computer Networks Laboratory	Gives practical exposure to network setup, protocols, and troubleshooting	SDG 9, 11
III-I	CS505PC	DevOps Laboratory	Builds practical deployment, CI/CD, automation, and systems integration skills	SDG 9, 12
III-I	AE506HS	Advanced English Communication Skills Laboratory	Improves technical communication, presentation, and teamwork in projects	SDG 4
III-I	CS507PC	Skill Development Course / UI Design – Flutter	Supports user-focused app design, prototyping, and mobile solution building	SDG 4, 9
III-II	CS601PC	Machine Learning	Helps students build predictive and data-driven intelligent solutions	SDG 3, 4, 9, 11
III-II	CS602PC	Formal Languages and Automata Theory	Builds strong analytical foundation for computing systems and compilers	SDG 4, 9
III-II	CS603PC	Artificial Intelligence	Enables intelligent problem solving and decision support systems	SDG 3, 4, 9, 11
III-II	CS604PC	Machine Learning Laboratory	Applies ML models to real datasets and practical prediction problems	SDG 3, 4, 9, 11
III-II	CS605PC	Artificial Intelligence Laboratory	Builds working AI solutions for classification, optimization, and reasoning	SDG 3, 4, 9, 11

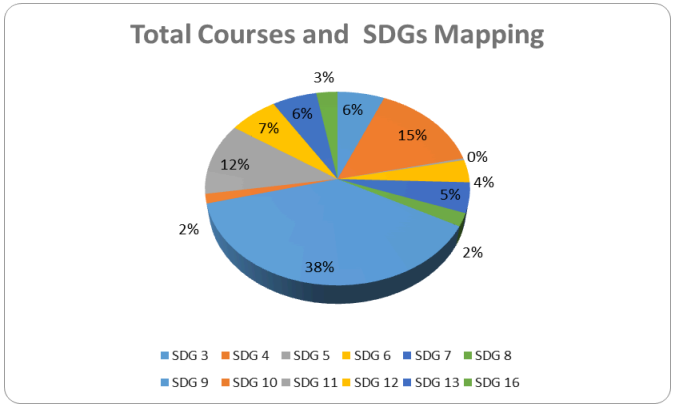
III-II	Professional Elective-III Lab	Professional Elective-III Laboratory	Provides hands-on implementation in emerging technologies	Depends on elective chosen
III-II	CS606PW	Industrial Oriented Mini Project / Internship / Skill Development Course (Big Data – Spark)	Gives industry exposure and real-world problem solving under practical constraints	SDG 8, 9, 12
III-II	ES610MC*	Environmental Science*	Reinforces sustainability, environmental protection, and responsible technology use	SDG 6, 12, 13
IV-I	CS701PC	Cryptography and Network Security	Addresses secure systems, privacy, and trustworthy digital infrastructure	SDG 9, 16
IV-I	CS702PC	Compiler Design	Builds system-level design thinking, language processing, and optimization ability	SDG 9
IV-I	CS704PC	Cryptography and Network Security Laboratory	Applies security concepts to real systems and practical implementation	SDG 9, 16
IV-I	CS705PC	Compiler Design Laboratory	Builds hands-on experience in language tools, parsing, and system design	SDG 9
IV-I	Project Stage–I (PW)	Project Stage–I	Focuses on problem identification, literature survey, requirement analysis, and design planning	Depends on project theme
IV-II	MB801BS	Organizational Behavior	Supports teamwork, leadership, stakeholder handling, and project coordination	SDG 8, 16
IV-II	CS802PW	Project Stage–II, including Seminar	Full implementation, testing, validation, documentation, and presentation of capstone solution	Depends on project theme

Total No. of Courses mapped to SDGs:

SDG	Name of SDG	Total Courses Mapped
SDG 3	Good Health and Well Being	20
SDG 4	Quality Education	48
SDG 5	Gender Equaity	1

SDG 6	Clean Water and Sustainability	12
SDG 7	Affordable and Clean Energy	16
SDG 8	Decent Work and Economic Growth	7
SDG 9	Industry, Innovation and Infrastructure	121
SDG 10	Reduced Inequalities	5
SDG 11	Sustainable Cities and Communities	38
SDG 12	Responsible Consumption and Production	21
SDG 13	Climate Change	19
SDG 16	Peace, Justice, and Strong Intitutions	9

Graph Representing the Total No. of Courses mapped to SDGs:

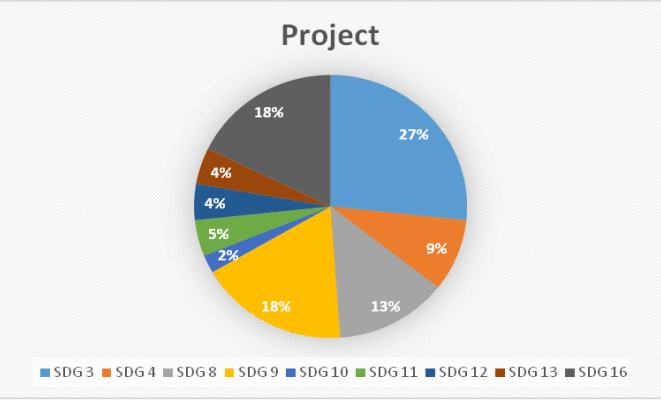


Total No. of Projects mapped to SDGs:

SDG	Count
SDG 2	3
SDG 3	12
SDG 4	4
SDG 8	6
SDG 9	8
SDG 10	1
SDG 11	2

SDG	Count
SDG 12	2
SDG 13	2
SDG 16	8

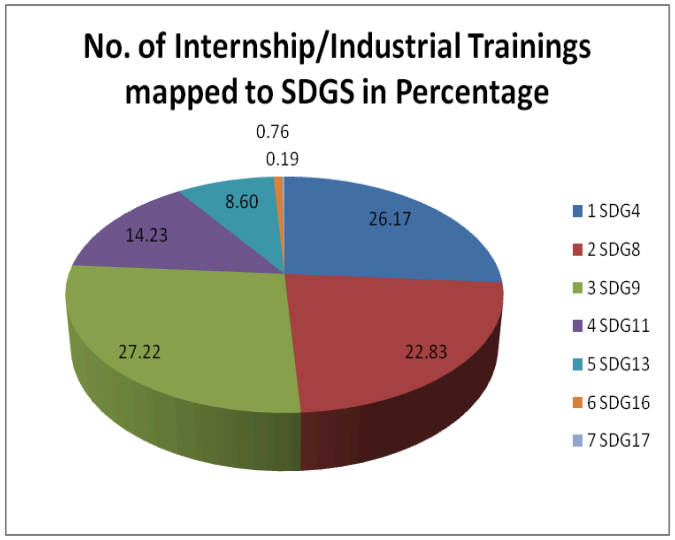
Graph Representing the No. of Projects mapped to SDGs:



Total No. of Internship/Industrial Trainings **mapped to SDGs:**

SNO	Mapped SDGs	No. of Internship/Industrial Trainings
1	SDG4	274
2	SDG8	239
3	SDG9	285
4	SDG11	149
5	SDG13	90
6	SDG16	8
7	SDG17	2

Graph Representing the No. of Projects mapped to SDGs:



2.8 Steps Taken for Enhancing Industry Institute Partnerships (15) — **Total Marks 15.00**

Institute Marks : 15.00

Industry Institute Partnerships

Anurag Engineering College signed Memorandum of Understanding (MOU) with several industries to establish official partnerships to bridge the gap between the academy and industry needs and thereby develop employability skills of the students.

Anurag Institution conducts training programs through MOUs with TASK, Oracle Academy, ALTAIR,TGIC,IBSC,Soham Academy,e-Stamp,IMARTICUS Learning and Sushmma Hegde offers skill programs, training program to provide awareness on trends and technology to enhance the employability skills.

List of MoUs

S.No	Name of the Company	Year of Signing MOU	Duration
1	TASK	08.09.2025	1 Year
2	Oracle Academy	29.12.2025	2 Years
3	ALTAIR	09.01.2024	1 Year
4	TGIC	26.03.2025	1 Year
5	IBSC	16.02.2026	1 Year

6	Soham Academy	01.02.2024	1 Year
7	e-Stamp	08.05.2025	5 Years
8	Elbert Technologies	.6.01.2025	1 Year
9	IMARTICUS Learning	04.11.2024	2 Years
10	Sushmma Hegde	25.09.2025	1 Year

Sample MoUs



Memorandum of Understanding

Innovation and entrepreneurship are key drivers of economic growth and social progress. To facilitate grassroots innovation and support early-stage startups, the Telangana Innovation Cell (TGIC), a unit of the Information Technology, Electronics & Communications Department, Government of Telangana, is launching the TGIC Host Institution Initiative. This initiative is designed to enable and support the higher educational institutions across Telangana in activating the local innovation ecosystem.

This **Memorandum of Understanding (the "Agreement")**, including the Annexures attached herewith and any amendments (collectively, the "Agreement") is entered on day of 25th March 2025, by and between:

TGIC Foundation (herein referred to as Telangana Innovation Cell) is an initiative by the Information Technology, Electronics & Communications Department, Government of Telangana. Telangana Innovation Cell was set up in 2017 under the State Innovation Policy with a vision - to build and nurture the Culture of Innovation in the State of Telangana which expression shall mean and include unless repugnant to the context, its successors, and assigns) of the One Part.

And

Anurag Engineering College and having its registered office at Ananthagiri (V & M), Kodad, Suryapet District, Telangana – 508206 (hereinafter referred to as 'Service Receiver' which expression shall, unless repugnant to the subject or context thereof, mean and include its successors and permitted assigns) of the OTHER PART.

The **TGIC Foundation** and **Anurag Engineering College** and shall hereinafter be referred to individually as a 'Party' and collectively as 'The Parties'.

- WHEREAS the **TGIC Foundation**, acts as the first response of the State Government for innovators, start-ups, entrepreneurs, and innovation catalysts to foster an inclusive innovation ecosystem with State-as-a-whole-approach.
- WHEREAS **Anurag Engineering College** is a higher educational institute is approved by AICTE New Delhi and affiliated to JNTU Hyderabad.



MEMORANDUM OF UNDERSTANDING (MoU)

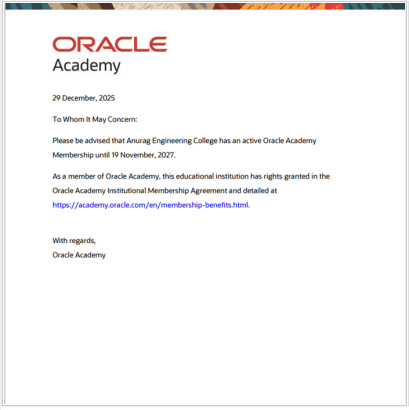
Between

IBSC Society

having its office at AMTZ Campus, Pragati Maidan, VM Steel Project S.O., Visakhapatnam – 530031, Andhra Pradesh, India
(First party, hereinafter referred to as IBSS),
Represented by Dr. Jitendra Kumar Sharma, Non – Executive President, IBSC Society & Managing Director & Founder CEO, AMTZ

And

Anurag Engineering College,
having its office at Ananthagiri (V & M), Kodad, Telangana
(Second party, hereinafter referred to as ANRK")
Represented by Dr. Suresh Kumar Tummala
Principal, Anurag Engineering College
Ananthagiri (V & M), Kodad, Telangana



ORACLE Academy

29 December, 2025

To Whom It May Concern:

Please be advised that Anurag Engineering College has an active Oracle Academy Membership until 19 November, 2027.

As a member of Oracle Academy, this educational institution has rights granted in the Oracle Academy Institutional Membership Agreement and detailed at https://academy.oracle.com/en/membership-benefits.html.

With regards,
Oracle Academy



ALTAIR

Request Altair RapidMiner - Data Analytics and AI platform

Academic Lab License

Our Institution would like to have Altair RapidMiner – Data Analytics and AI Platform as part of curriculum requirements and/or wants to use the Altair Software tools to teach students as part of lab component, hence request to issue Academic lab license to our college (Free).

College Name	ANURAG Engineering College
Name of Faculty	G. Srinivasa Rao
Mobile No	9553122275
Email ID (College domain Email ex: name@college.ac.in or *.edu)	hod.cse@anurag.ac.in
Branch of Engineering	Computer Science & Engineering
Number of License? (Equal to the no. of Systems in the Lab)	We request 70 User License for our lab

Signed in Agreement:

The Altair will issue Academic lab license which is commercial grade: **Altair RapidMiner – Data Analytics and AI Platform.** All these licenses need to be used only for academic purpose. Should not be used for commercial activities and for executing projects and services leading to revenue generation. Institution is entitled to generate revenue through teaching courses to students which are academic in nature.

This Letter of Intent will be operative for a period of 12 months from the date of signing or until superseded by a new agreement signed within the Letter of Intent period. We also agree that the students are exposed online Learning and certification courses for further competency building. We acknowledge in our website / brochures / online content that we use **Altair RapidMiner** for Data Analytics AI and ML Solutions as part of Teaching / Lab component. We give consent to Altair to use Our logo for Academic Collaboration disclosures.

Sign: Principal (or HOD) with Seal
(Authorized Signatory from Institution)
Date: 08-01-2024

Name: Dr. Suresh Kumar Tummala

Email: principal@anurag.ac.in

Mobile: 9553122270

Principal
Anurag Engineering College
Ananthagiri(V&M), Kodad
Suryapet Dt, Telangana-508206

Data Scientist Faculty in charge

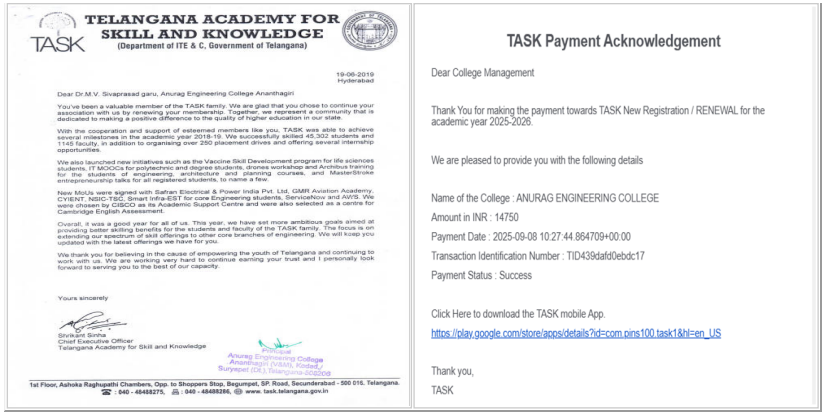
Date: 08-01-2024

Name: G. Srinivasa Rao

Email: hod.cse@anurag.ac.in

Mobile: 9553122275

Altair India Academic Initiative



TELANGANA ACADEMY FOR SKILL AND KNOWLEDGE
(Department of ITE & C, Government of Telangana)

TASK

19-06-2019
Hyderabad

Dear Dr.M.V. Sivaprasad garu, Anurag Engineering College Ananthagiri

You've been a valuable member of the TASK family. We are glad that you chose to continue your association with us by renewing your membership. Together, we represent a community that is dedicated to making a positive difference to the quality of higher education in our state.

With the cooperation and support of esteemed members like you, TASK was able to achieve several milestones in the academic year 2018-19. We successfully skilled 45,302 students and 1145 faculty, in addition to organising over 250 placement drives and offering several internship opportunities.

We also launched new initiatives such as the Vaccine Skill Development program for life sciences students, IT MOOCs for polytechnic and degree students, drones workshop and Architrus training for the students of engineering, architecture and planning courses, and MasterStroke entrepreneurship talks for all registered students, to name a few.

New MoUs were signed with Safran Electrical & Power India Pvt. Ltd, GMR Aviation Academy, CVIENT, NSIC-TSC, Smart Infra-EST for core Engineering students, ServiceNow and AVIS. We were chosen by CISCO as its Academic Support Centre and were also selected as a centre for Cambridge English Assessment.

Overall, it was a good year for all of us. This year, we have set more ambitious goals aimed at providing better skilling benefits for the students and faculty of the TASK family. The focus is on extending our spectrum of skill offerings to other core branches of engineering. We will keep you updated with the latest offerings we have for you.

We thank you for believing in the cause of empowering the youth of Telangana and continuing to work with us. We are working very hard to continue earning your trust and I personally look forward to serving you to the best of our capacity.

Yours sincerely

Shrikant Sinha,
Chief Executive Officer
Telangana Academy for Skill and Knowledge

Principal
Anurag Engineering College
Ananthagiri (VASM), Kodad,
Suryapet (Dt.) Telangana-508206

1st Floor, Ashoka Raghupathi Chambers, Opp. to Shoppers Stop, Begumpet, SP. Road, Secunderabad - 500 016. Telangana.
☎ : 040 - 48488275, 🖷 : 040 - 48488286, @ www. task.telangana.gov.in

TASK Payment Acknowledgement

Dear College Management

Thank You for making the payment towards TASK New Registration / RENEWAL for the academic year 2025-2026.

We are pleased to provide you with the following details

Name of the College : ANURAG ENGINEERING COLLEGE
Amount in INR : 14750
Payment Date : 2025-09-08 10:27:44.864709+00:00
Transaction Identification Number : TID439dafd0ebdc17
Payment Status : Success

Click Here to download the TASK mobile App.
https://play.google.com/store/apps/details?id=com.pins100.task1&hl=en_US

Thank you,
TASK

2024-2025

List of training programs:

S.No.	Date	Event
1	02.08.2024 to 06.08.2024	Aptitude and Reasoning training program for All the IV B.Tech students registered with TASK
2	23.09.2024 to 25.09.2024	Database Programming with SQL training program for IV B.Tech CSE students registered with TASK
3	25.09.2024	Gate Awareness Program for IV Year B.Tech students(online mode)
4	26.09.2024 to 28.09.2024	Programming Essentials in Python training program for All the III B.Tech students registered with TASK
5	21.10.2024 to 26.10.2024	C & Data Structures training program for IV B.Tech students registered with TASK
6	24.10.2024	AI Yatra is a part of the Telangana Vaibhavam AI Program, a collaborative initiative by Swecha, the Government of Telangana, TASK, IIIT Hyderabad, Ozonetel,

		and Tech Vedika for All B.Tech students
7	11.11.2024 to 13.11.2024	Cloud Computing training program training program for IV B.Tech CSE students registered with TASK
8	20.03.2025	LinkedIn Career Kickstarter Program for IInd,IIIrd & IV Year B.Tech students
9	20.03.2025	Gate Awareness Program for IIIrd Year B.Tech students
10	10.04.2025	OVERSEAS EDUCATION AWARENESS PROGRAM for Final Year B.Tech students

Sample document related to Programs:



ANURAG Engineering College
Autonomous
Ananthagiri (V&M), Suryapet (Dt.)

Date: 07.07.2023

CIRCULAR

All the IV B.Tech students registered with TASK are informed to attend the training program on **Aptitude and Reasoning** scheduled from 10.07.2023 and 11.07.2023.

Time : 09:30AM to 1:00PM & 01:30PM to 4:00PM

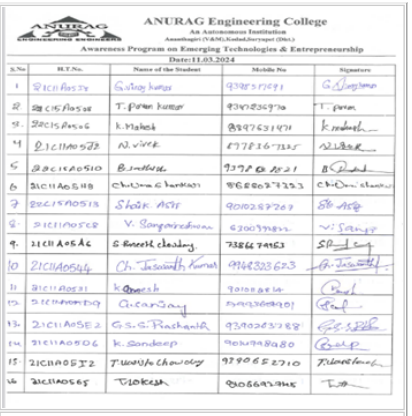
Venue : Seminar Hall, Block – D

- *Carrying mobile phones is strictly prohibited.*

TPO

Copy To
All the HoDs & TPO coordinators to circulate among their IV B.Tech students.
Dept. notice boards.

PRINCIPAL
Principal
Anurag Engineering College
Ananthagiri(V&M), Kodad
Suryapet Dt. Telangana 508206



ANURAG Engineering College
An Autonomous Institution
Ananthagiri (V&M),Kodad,Suryapet (Dist.)
Awareness Program on Emerging Technologies & Entrepreneurship
Date:11.03.2024

S.No	H.T.No.	Name of the Student	Mobile No	Signature
1	21C11A05J4	Gutlaj kumar	9398571491	G.Sanjeeva
2	22C15A0558	T. Pavan kumar	9347835970	T. Pavan
3	22C15A0546	k.Mahesh	8897631471	K.mahesh
4	21C11A0573	N.vivek	6778167335	N.Vivek
5	22C1KA0510	B.sathvik	9398 [illegible] 21	B.Sathvik
6	21C11A05H8	Chitrangi harikiran	8688027323	Chitrangi harikiran
7	22C15A0513	Shaik. Asif	9010287267	Shaik. Asif
8	21C11A0558	V. Sangameshwar	6300379832	V. Sangameshwar
9	21C11A05A6	S. Suresh chandra	7386674953	S. Suresh chandra
10	21C11A0544	Ch. Jaswanth, Kumar	9948333623	Ch. Jaswanth, Kumar
11	21C11A0531	k.Ramesh	9010886614	K.Ramesh
12	21C11A0559	G.Karthik	8499368491	G.Karthik
13	21C11A05E2	G.S.S.Prashanth	9390263788	G.S.S.Prashanth
14	21C11A0506	k.Sandeep	9014948480	K.Sandeep
15	21C11A05I2	T.Venkata Chowdary	9390652710	T.Venkata Chowdary
16	21C11A0565	T.Lokesh	9205692348	T.Lokesh



ORACLE
Academy

AWARD of COURSE COMPLETION

Java Fundamentals

PRESENTED TO

GATTU VIGNESH

FOR SATISFACTORY COMPLETION OF ALL COURSEWORK

31st March 2023

William McCabe
Vice President, Oracle Academy



ORACLE
Academy

AWARD of FINAL EXAM COMPLETION

Java Fundamentals

PRESENTED TO

SRI HAINDAVI PICHIKALA

FOR SATISFACTORY COMPLETION OF COURSE FINAL EXAM

3rd March 2023

William McCabe
Vice President, Oracle Academy


GPS Map Camera
Anantha Giri, Telangana, India
Ananthagiri (v), Kodad (m Nalgonda (distirict, Ananthagiri Rd,
Anantha Giri, Telangana 508206, India
Lat 17.039929° Long 79.976712°
16/10/2025 09:57 AM GMT +05:30
Google


GPS Map Camera
Anantha Giri, Telangana, India
Ananthagiri (v), Kodad (m Nalgonda (district, Ananthagiri Rd, Anantha Giri,
Telangana 508206, India
Lat 17.039842° Long 79.976707°
17/10/2025 09:41 AM GMT +05:30
Google